

Remember What You Thought, Not What You Said: Self-Distilled Memory Admission for LLM Agents

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Abstract

LLM agents must choose memories before future questions arrive, although useful concepts are often unstated inferences. Importance judgments favor candidates the passage states: the importance report distinguishes useful concepts among unstated candidates alone but falls below chance on the standard pool (the generator’s original candidate lists), where useful concepts are usually unstated. The model’s workspace readout (a candidate’s peak reciprocal rank under the logit lens across layers) ranks useful concepts above chance on this pool. Self-Distilled Admission (SDA) labels the top two candidates by the workspace readout and output likelihood and, through the model’s importance prompt, trains a LoRA adapter that is used only for admission requests. Gradient updates use only labels the frozen model generates, with no future question or reward. Across six models, SDA improves answer accuracy over each model’s own report in 54 of 54 model–seed–budget comparisons on the primary statedness-balanced pool. Qwen3-8B gains 10.7 points at two stored concepts; published admission criteria (ConsistencyGate, A-MAC, Nemori) score below the untrained report. An off-the-shelf serving engine runs the adapter at 242 candidates per second. The model internally encodes concepts worth storing that its importance report does not reflect; SDA trains the model’s answer to its importance prompt to admit those concepts.

1 Introduction

An LLM agent must choose what to remember before it knows the next question, and the useful concept is often an inference that the passage never states. A description of a landmark can identify a city without naming it, so remembering that city preserves the link to a later question about the local language. Once the passage leaves the context, the agent’s answer depends on which inferences survived admission to memory.

Memory systems make this choice through importance judgments: the model’s importance report, its 1–10 importance rating (Park et al., 2023), or published admission criteria built on context support (Zhang et al., 2026; Zhang and Li, 2026). In each case, the passage provides the evidence for admission, but an agent also needs to preserve what it inferred from the passage. This creates a gap between recognizing an inference during reading and selecting it for later use: a memory system can discard the useful concept even when the model has identified it.

The importance report is the model’s yes/no answer to whether a candidate is among the most important things to remember, scored by $P(\text{yes})$ against $P(\text{no})$. Controlled pools fix each episode’s useful concept and vary which candidates are stated (present in the passage), so a drop in area under the ROC curve (AUC) isolates the effect of statedness. The standard pool uses the generator’s original candidate lists, where the useful concept is usually the only unstated candidate; in the statedness-balanced pool, the passage states the useful concept in half the episodes. The report’s pooled AUC falls from 0.880–0.897 among unstated candidates alone to 0.658–0.736 on the statedness-balanced pool and 0.322–0.427 on the standard pool, ranking useful concepts below chance against stated distractors. The 1–10 importance rating and published criteria built on context support also favor stated candidates.

The model’s workspace readout, the peak reciprocal rank of a candidate under the logit lens across layers, ranks the useful concept above chance on the same standard pool. The model’s intermediate layers therefore encode information that its importance report does not express. Naming the useful concept in an otherwise unchanged episode raises AUC for both signals, with a larger increase for the workspace readout (Figure 1).

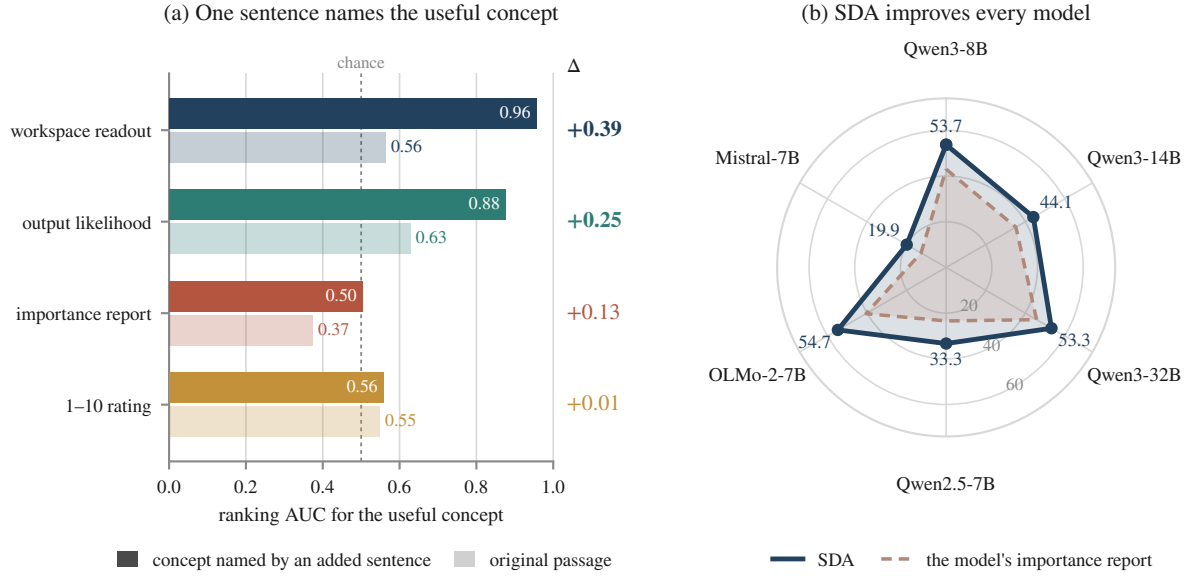


Figure 1: Naming the useful concept raises AUC far more for the workspace readout and output likelihood than for the importance report and 1–10 importance rating; Self-Distilled Admission (SDA) distills these two rankings into admission decisions. (a) Six-model mean pooled AUC on the standard pool: original passage (pale) or with one added sentence naming the useful concept (solid); dashed line: chance. (b) Answer accuracy (%) at two stored concepts on the statedness-balanced pool: SDA (solid, three-seed mean) against each model’s report (dashed); mean gain 9.8 points.

Self-Distilled Admission (SDA) transfers the model’s candidate rankings into its admission decision (Figure 3). The teacher is a frozen model’s ranking of each episode’s candidates by workspace readout and output likelihood (the candidate’s mean token log-probability given the passage); SDA labels the top two for admission and trains a low-rank adaptation (LoRA) adapter (Hu et al., 2021) on those labels through the model’s importance prompt. Optimization uses only the passage, candidates, and labels generated by the model.

SDA improves answer accuracy over existing memory systems’ importance judgments. On the primary statedness-balanced pool, it exceeds each model’s own importance report in 54 of 54 comparisons across models, seeds, and memory budgets, and exceeds the 1–10 importance rating in 52 of 54 (Table 1). With two stored concepts, Qwen3-8B gains 10.7 accuracy points over its importance report and 5.2 over the rating. ConsistencyGate, A-MAC, and Nemori fall below the untrained report. SDA exceeds these published admission criteria by 14.1–27.8 points at the same memory budget (Table 2).

The paired-passage set, in which two episodes contain the same candidates in different orders but differ in passage, question, and useful concept, tests

the teacher’s use of the passage because only the passage identifies the useful concept. Across five models, removing the workspace readout from the teacher lowers answer accuracy by 1.9 points at one stored concept (Table 4).

SDA is a plug-and-play adapter that an off-the-shelf serving engine applies only to admission requests. On the standard pool, the served Qwen3-8B adapter keeps 50.1% answer accuracy at 242 candidates per second (Table 5).

Our contributions are:

- **Stated-content preference in importance judgments.** Importance judgments order candidates by statedness, a preference that controlled pools separate from usefulness, while the workspace readout tracks unstated useful concepts (§6.1).
- **Plug-and-play SDA module.** SDA distills the workspace readout and output likelihood into the model’s importance prompt through an adapter that serves only admission requests (§5).
- **Results across models and memory settings.** Evaluation across six models from three families, three seeds, three memory budgets, and

five candidate pools establishes answer accuracy gains over the importance report, the rating, and published admission criteria, while ablations on the paired-passage set show that both teacher signals contribute to answer accuracy (§6.2, §6.4).

2 Background and Related Work

Memory systems and evaluation. Generative Agents combines ratings of importance with recency and relevance for retrieval (Park et al., 2023). MemoryBank, MemGPT, Mem0 and A-MEM provide persistence, memory operations and organization (Zhong et al., 2024; Packer et al., 2023; Chhikara et al., 2025; Xu et al., 2025). LoCoMo and LongMemEval evaluate memory over long conversations (Maharana et al., 2024; Wu et al., 2025).

Admission criteria. ConsistencyGate filters candidates by self-consistent support in the source context (Zhang and Li, 2026), while A-MAC combines LLM assessment with textual support and memory attributes (Zhang et al., 2026). A multi-factor value model optimizes evidence retention (Chen and Cheng, 2026), and Causal Memory Intervention scores answers to a known request under memory interventions (Srivastava, 2026). SDA instead draws targets from the frozen model’s own workspace readout and output likelihood.

Learned memory management. Nemori distills interaction sequences into memory using prediction error (Ma et al., 2026). MemCon learns a policy for retrieval, consolidation and forgetting (Jiang et al., 2026), while AgeMem trains memory tool actions with reinforcement learning (Yu et al., 2026). The survey of Hu et al. (2025) situates these approaches within memory systems.

Internal signals and representation decoding. EM-LLM, SirLLM and Titans use surprisal, entropy and gradient-based surprise for memory (Fountas et al., 2025; Yao et al., 2024; Behrouz et al., 2025). The logit lens, a J-lens with the identity map, decodes internal representations (nostalgabraist, 2020; Gurnee et al., 2026), as do tuned lenses, Future Lens and Patchscopes (Belrose et al., 2023; Pal et al., 2023; Ghandeharioun et al., 2024). SDA’s workspace readout applies the logit lens across layers (§5.1).

Memory distillation and self-distillation. Knowledge distillation transfers teacher predic-

tions to a student, including between models of the same architecture (Hinton et al., 2015; Furlanello et al., 2018). TMEM writes distilled experience into low-rank weights (Ren et al., 2026), while Agent Memory Distillation transfers a larger teacher agent’s trajectories into structured memory for a smaller student (Kim et al., 2026). Unlike TMEM and Agent Memory Distillation, we use the same model as teacher and student.

Internal information and model reports. Internal states can predict correctness beyond expressed confidence (Kadavath et al., 2022; Azaria and Mitchell, 2023; Orgad et al., 2025). Stated reasoning can omit influences on an answer (Turpin et al., 2023), and salience can be poorly reported (Tienes et al., 2025). Introspection training teaches models to predict their behavior (Binder et al., 2025).

3 The Admission Setting

Our setting makes one useful concept compete with passage concepts for a fixed number of memory slots. Each episode contains a passage, an ordered candidate set, utility labels, and a future question with its answer (Figure 2). With two memory slots, the city in Figure 2 competes with the elevator, gift shop, and scenery. Only the passage and candidates reach the admission policy.

3.1 Episodes

Each episode centers on a bridge: information linking the passage to a future answer. **Decoupled** requires two inference steps: recover the bridge from the passage, then combine it with general knowledge to answer the future question. GPT-4.1 generates a passage about a named person, together with the bridge, question, and answer. The answer is excluded from the passage and candidate set, while the question connects to the passage through the person’s name.

The generator assigns gold utility labels using the bridge and the question it writes: positive for the bridge, which we call the episode’s useful concept, and negative for vivid surface distractors and neutral passage concepts. Rule-based filters check candidate strings, bridge and answer exclusion, and question overlap, while deduplication removes repeated concepts. A blind alias audit checks that recovering the bridge still leaves an inference to the answer (App. A).



Figure 2: Memory admission within an episode. The policy receives a passage and candidate concepts before the future question arrives and stores k concepts. A frozen reader later answers the question from those concepts alone. Here, the useful concept, *dubai*, is the only candidate the passage never states. Figure 5 gives the episode in full. *Drawing guide* (6.3 in $\times$ 2.1 in, Figure 1 palette): five stages left to right, joined by thin arrows. (1) Passage card with two or three lines of Jibril’s passage, “Burj Khalifa” lightly highlighted. (2) Six candidate chips: *elevator*, *desert*, *gift*, *observation and tourists* in grey (stated); *dubai* in light blue (unstated) with a gold star (useful); a small legend. (3) An “Admission policy” box feeding a memory with $k = 2$ slots that hold *dubai* and *elevator*. (4) A dashed vertical line labeled “later”, then a question bubble: “What language was Jibril most likely to hear from local staff?” (5) A frozen reader (grey box with a snowflake) that reads only the two memory slots and answers “arabic”. Small grey notes under the arrows: “policy sees passage and candidates; question hidden” and “reader sees the question and memory only”.

Explicit states the useful concept in the passage. **Compositional** requires a multi-hop answer and permits passage anchors in the question. The **interim held-out set** and **confirmation set** repeat the Decoupled construction with new passages and bridges absent from the training episodes. The confirmation set also excludes the interim held-out set’s contexts and is locked after configuration selection. Each set is locked before evaluation.

3.2 Candidate Pools and Controls

The pools and controls separate the effects of passage mention, candidate competition, context, generator, and bridge category. The **standard pool** is the generator’s candidate list for each confirmation-set episode: 1,607 candidates, 4.43 per episode. Each pool and control preserves each episode’s future question, answer, and original utility labels; added candidates receive negative labels.

Mention and position. These controls isolate the named concept and sentence position. The **stated-bridge control** appends *The passage also mentions bridge*. The **position control** places that sentence before the passage, whereas the **distractor-named control** substitutes a distractor for the bridge in the appended sentence.

Competition for memory. Our primary **statedness-balanced pool** uses the 363 held-out confirmation-set episodes in fixed order. Odd positions, counted from the first episode, keep the passage unchanged; even positions append *The passage also mentions bridge*. with *bridge* replaced by the useful concept. Under the case-insensitive stem rule, the stated-concept condition contains 190 episodes: all even positions and 9 odd positions whose original passages already match the useful concept. The unstated-concept condition contains 173 episodes, all at odd positions.

Every episode retains its original candidates, including stated distractors, and adds the first four (odd positions) or three (even positions) additional unstated candidates, in order, from the pool with unstated distractors described below. Position alone determines the naming sentence and distractor count, with no sampling (App. A.4).

The resulting 363 episodes average 7.93 candidates, with useful concepts accounting for 13.3% of stated candidates and 12.0% of unstated candidates. A statedness-only rule ranks near chance, with pooled AUC of 0.515 and within-episode AUC of 0.516.

The **naming-balanced pool** names four candidates in every episode using five sentence templates

at varying positions; named candidates are useful 12.5% of the time, against a 12.6% base rate.

Each pool balances one cue and leaves the other imbalanced in the opposite direction: in the statedness-balanced pool, the added naming sentence always names the useful concept, whereas in the naming-balanced pool, stated candidates are more often distractors.

Two further pools start from the standard pool, increasing the alternatives admission must rank for the same memory budget. The **pool with unstated distractors** adds four confusable concepts from the bridge’s semantic category. An LLM proposes these concepts, which are filtered for absence from the passage and original pool. The **enlarged pool** adds further concepts from other episodes, reaching 12.41 candidates on average.

Context, generator, and bridge category. These controls test whether admission depends on the episode’s own passage, the generator, or the kind of useful concept. **Passage replacement** replaces each passage with another episode’s passage while retaining its candidates and labels. The **paired-passage set** pairs confirmation-set episodes that contain the same candidates in different orders while each retains its passage, future question, and useful concept (§6.4, App. H.1). The **alternate-generator set** uses GPT-4o to write episodes with an unstated useful concept. The **entity-bridge set** uses GPT-5.6 to place a named-entity bridge among scene-object distractors (App. G.3).

4 Evaluation Setup

Given a passage x and ordered candidates $C = (c_1, \dots, c_m)$, admission stores $K \subseteq C$ with $|K| \leq k$. The future question q and answer a are hidden. When q arrives, the frozen reader sees only q and K ; the original passage has left its window. Every policy receives the same candidates and budget. We measure three successive outcomes: within-episode ranking, useful-concept retention, and answer accuracy.

Within-episode ranking. Within-episode area under the receiver operating characteristic curve (AUC) measures how well a policy ranks candidates competing for memory. It is the fraction of pairs of the useful concept and another candidate ordered correctly, with half credit for ties, averaged across episodes. For useful concept b_i in episode i ,

score s and n episodes,

$$A_{\text{episode}} = \frac{1}{n} \sum_{i=1}^n \frac{1}{|C_i| - 1} \times \sum_{c \in C_i \setminus \{b_i\}} \left[\mathbf{1}\{s(b_i) > s(c)\} + \frac{1}{2} \mathbf{1}\{s(b_i) = s(c)\} \right]. \quad (1)$$

Pooled AUC compares candidates across the entire set and summarizes score discrimination.

A candidate is stated when a case-insensitive stem match finds it in the passage and unstated otherwise. Replacing utility labels with statedness labels gives the *statedness AUC*.

Useful-concept retention. Useful-concept retention B_k is the fraction of episodes whose stored set contains the useful concept:

$$B_k = \frac{1}{n} \sum_{i=1}^n \mathbf{1}\{b_i \in K_i\}. \quad (2)$$

A policy can rank it above most distractors and still omit it at a small budget. Gold-concept selection reads the gold utility label to rank the labeled useful concept first and appears as a reference row in the tables.

Answer accuracy. Answer accuracy is the fraction of future questions that a fixed, frozen reader answers correctly from the stored concepts under blind semantic grading. Accuracy also depends on the reader and the other stored concepts, so retention and accuracy can differ.

The reader answers greedily, with Qwen3 thinking disabled. A blind semantic judge receives only the question, gold answer and response. It accepts a committed correct answer or exact equivalent. A separate string-match grader checks agreement (App. D.1).

The closed-book reader (question only) and the full-context reader (whole passage) measure answer recovery before admission and with the complete original input, respectively.

Comparisons. Baselines include the importance report, a 1–10 importance rating, the reversed importance report, unstated-first (unstated candidates first, ties broken at random), unstated-first with report ordering (unstated candidates first, each statedness class ordered by the importance report), implied-content wording (an importance

prompt that permits implied content), statedness-cued wording (an importance prompt that says the passage may never mention the useful concept), and the statedness-adjusted report (the importance report plus one fitted offset for stated candidates). We also compare the combined teacher used directly, random selection, output likelihood, the workspace readout, and adapters (students, §5.2) trained on labels from one teacher signal (App. B.4).

Trained results are the mean and sample standard deviation over three seeds; differences between policies are differences of means (Table 14). Bold table entries identify SDA.

Evaluation scale. The main diagnostic comparison covers eight models from three families: Qwen2.5-7B, OLMo-2-7B, Mistral-7B and Qwen3 at 1.7B, 4B, 8B, 14B and 32B (Qwen Team, 2025; Walsh et al., 2025; Jiang et al., 2023; Yang et al., 2025). Outside the checkpoint grid of App. G.2, the short names Qwen2.5-7B, OLMo-2-7B and Mistral-7B denote the instruct checkpoints Qwen2.5-7B-Instruct, OLMo-2-1124-7B-Instruct and Mistral-7B-Instruct-v0.3. The extended grid covers 23 base and post-trained checkpoints from these families and GPT-2 (App. G.2).

SDA training covers six models with three seeds each: Qwen3-8B, Qwen3-14B, Qwen3-32B, Qwen2.5-7B, OLMo-2-7B and Mistral-7B, yielding 18 training runs. Each model supplies its own teacher, adapter and frozen reader.

Training and validation splits contain 175 and 45 episodes, respectively, from 88 Explicit and 132 Implied-answer episodes, including the 57-episode alternate-generator set. In an Implied-answer episode, the passage cues an unstated useful concept that supplies the answer.

Development uses Decoupled (68 episodes), Compositional (52), and two separate 68-episode sets adding confusable or unrelated unstated distractors to Decoupled. The interim held-out set and the confirmation set contain 316 and 363 episodes, respectively, while the entity-bridge set contains 42 episodes. Passage controls and pools on the confirmation set reuse its 363 episodes.

Evaluation uses memory budgets $k = 1, 2, 3$. The main comparison uses the statedness-balanced pool, reporting $k = 2$ for all six models. On the confirmation set, both graders score all 13,068 answers of the 18 SDA-report comparisons.

5 Self-Distilled Admission

SDA distills a frozen model’s candidate rankings into its answer to the importance prompt (Figure 3; Algorithm 1).

5.1 Teacher Rankings

The teacher ranks each episode’s candidates using two signals from the frozen model, defined next.

Workspace readout. The workspace readout measures how readily the unembedding recovers a concept during context processing. SDA uses only the logit lens, which decodes each layer’s residual state through final normalization and unembedding and is the J-lens with $J = I$ (nostalgebraist, 2020; Gurnee et al., 2026). Let $h_\ell(x) \in \mathbb{R}^d$ be the residual state at the final context position after transformer layer ℓ , with L layers in total. Final normalization LN and unembedding W_U give vocabulary logits $z_\ell(x) = W_U \text{LN}(h_\ell(x))$. Writing $\text{rank}_\ell(t) = 1 + |\{v : z_{\ell,v} > z_{\ell,t}\}|$, we take the peak reciprocal rank across layers and first-token forms:

$$W(c; x) = \max_{\substack{1 \leq \ell \leq L \\ t \in T(c)}} \frac{1}{\text{rank}_\ell(t)}. \quad (3)$$

Here $T(c)$ contains the distinct first tokens of the candidate’s bare and leading-space forms, each in original and capitalized form. The maximum captures the candidate’s strongest decoding rank across these layers and forms. Computing the readout requires one forward pass over x , using final-position states from all layers.

Combined teacher. Output likelihood scores the complete candidate by its mean log probability per token: $\text{LL}(c; x) = \log p_{\mathcal{M}}(c \mid x) / |c|$, where $|c|$ is the candidate’s token length. To put workspace readout and output likelihood on a common scale, we map each to mid-rank percentiles within the episode. Tied values receive their mean rank (App. B). The teacher used in the main evaluation averages these percentiles:

$$T(c; x) = \frac{1}{2} (P_W(c; x) + P_{\text{LL}}(c; x)). \quad (4)$$

Here P_S denotes the percentile under score S . The two highest-ranked candidates receive the binary label $\tilde{y} = 1$ for *yes*; all others receive $\tilde{y} = 0$ for *no*. Candidate order breaks score ties. The frozen model generates every teacher label from the passage and candidates.



Figure 3: SDA. (a) The frozen model combines workspace readout and output likelihood to rank each episode’s candidates and labels the top two yes. (b) A LoRA adapter learns these labels through the model’s own importance prompt. (c) The serving engine applies the adapter only to admission requests. *Drawing guide (6.3 in $\times$ 2.45 in, Figure 1 palette): three panels. (a) Teacher: passage x and candidates C enter a frozen model (grey box with a snowflake). Beside it, a small heat strip with layers on the vertical axis (1 at the bottom, L at the top) and one column per candidate, shaded by the reciprocal rank of the candidate’s first token at that layer; a navy dot marks each column’s peak, labeled $W(c; x)$. The output head is labeled $LL(c; x)$ in teal. Both become within-episode percentile bars, averaged into T ; a sorted candidate list tags the top two yes (rust) and the rest no (grey). (b) Student: the same frozen model with a small rust LoRA block, answering the importance prompt “Is c one of the most important things to remember...? yes/no”; a dashed line to the labels of (a) reads “next-token cross-entropy”. (c) Deployment: a host agent sends passage and candidates to a serving engine (vLLM); admission requests run with the adapter on and store the top k in memory; question requests run with the adapter off, and the frozen reader answers.*

5.2 Admission Distillation

The student is a LoRA adapter on the frozen model that learns the teacher’s labels by answering the importance prompt.

Admission score. The importance prompt $u(c, x)$ asks whether c is one of the most important concepts to remember for possible future questions and requests a single yes or no (App. B.6). Qwen3 uses thinking-disabled prompts. Let Y, N contain the distinct first tokens of the case and leading-space forms of *yes* and *no*, respectively. The admission score compares their total probabilities:

$$s_{\theta}(c; x) = \log \sum_{t \in Y} p_{\theta}(t \mid u(c, x)) - \log \sum_{t \in N} p_{\theta}(t \mid u(c, x)), \quad (5)$$

$$V_{\theta}(c; x) = \sigma(s_{\theta}(c; x)).$$

The untrained model’s score V is the importance report; SDA ranks candidates by the adapter-enabled score s_{θ} .

Training objective. We optimize only the LoRA adapter; base parameters remain frozen. Cross-

entropy trains next-token prediction of the teacher’s target word:

$$\mathcal{L}(\theta) = -\frac{1}{|\mathcal{D}|} \sum_{(x, c) \in \mathcal{D}} \log p_{\theta}(\tau(\tilde{y}(c; x)) \mid u(c, x)). \quad (6)$$

The training examples in $\mathcal{D}$ pair each passage x with each of its candidates c and the corresponding teacher label $\tilde{y}(c; x)$. Here τ maps a positive label to *yes* and a negative label to *no*. The target probability is normalized over the vocabulary. Prompt tokens are masked from the loss (App. B.2).

Training and selection. We train a rank-16 adapter for two epochs and select the earliest checkpoint with the highest within-episode AUC on the in-distribution validation split. Optimization uses teacher labels; validation uses gold utility labels. We select the combined teacher by mean answer accuracy across development sets and training seeds, scored by the development run’s automatic grader. The configuration is fixed before evaluation on the confirmation set (App. C).

Algorithm 1 Self-Distilled Admission.

Require: frozen model $\mathcal{M}$; training episodes $\mathcal{E}$, each a passage x with candidates C

- 1: **for** each episode $(x, C) \in \mathcal{E}$ **do**
 - 2: **for** each candidate $c \in C$ **do**
 - 3: $W(c; x) \leftarrow$ peak reciprocal rank across layers $\triangleright$ Eq. (3)
 - 4: $LL(c; x) \leftarrow \log p_{\mathcal{M}}(c | x) / |c|$
 - 5: $T(c; x) \leftarrow \frac{1}{2}(P_W(c; x) + P_{LL}(c; x))$ for all $c \in C$ $\triangleright$ Eq. (4)
 - 6: $\tilde{y}(c; x) \leftarrow 1$ for the two highest T , else 0
 - 7: train adapter θ on $\mathcal{L}(\theta)$ over all $(x, c, \tilde{y})$ $\triangleright$ Eq. (6)
 - 8: select the earliest checkpoint with the highest validation within-episode AUC
 - Deployment:** given a passage x , candidates C and budget k
 - 9: store the k candidates with the highest $s_{\theta}(c; x) \triangleright$ Eq. (5)
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5.3 Off-the-Shelf Serving

At deployment, SDA makes one yes/no forward pass per candidate and reads the importance prompt’s output probabilities to compute s_{θ} . The host stores the top- k candidates and manages retrieval. Candidate order breaks score ties.

The adapter is attached only to admission requests in an off-the-shelf serving engine such as vLLM (Kwon et al., 2023). Prefix caching reuses the shared passage. All other requests, including the frozen reader’s answer generation, use the unmodified model.

6 Results

6.1 Statedness and Importance Judgments

The importance report ranks useful concepts well among unstated candidates but loses that ordering when stated and unstated candidates compete. Among unstated candidates, its pooled AUC is 0.880–0.897 across six models, compared with 0.592–0.697 for the workspace readout (Table 16). Report AUC falls to 0.658–0.736 on the statedness-balanced pool and below chance to 0.322–0.427 on the standard pool, where statedness AUC is 0.582–0.688.

The workspace readout ranks useful concepts higher than the importance report on the statedness-balanced pool, with pooled AUC 0.719–0.812. Statedness alone ranks candidates near chance on this pool (§3).

On the original standard-pool passages, the six-model mean workspace AUC is above chance at 0.563. Naming the useful concept raises it to 0.956 while report AUC rises from 0.373 to 0.504 (Table 20). Naming a distractor lowers both on Qwen3-8B (App. F.4). Statedness-cued wording also re-

sponds to naming, but Qwen3-8B AUC falls from 0.537 when the useful concept is unstated to 0.417 when named; it is 0.621 on Explicit (App. F.3).

In 95% of standard-pool episodes, the useful concept is the sole unstated candidate. At $k = 2$, unstated-first and unstated-first with report ordering both reach 53.4% answer accuracy; the reversed importance report reaches 46.6%. Balancing statedness lowers answer accuracy for all three policies because their preference for unstated candidates now also selects the added unstated distractors. All three then rank below the importance report at $k = 2, 3$; unstated-first with report ordering exceeds it at $k = 1$ (Tables 2, 13 and 14).

SDA trails unstated-first with report ordering on the standard pool by 9.5 points at one stored concept. At two stored concepts, the difference is -3.4 points. On the statedness-balanced pool, SDA exceeds this rule by 14.6 points at two stored concepts (Tables 14 and 8).

6.2 Admission Results

SDA exceeds each model’s own importance report in all 54 model–seed–budget comparisons on the statedness-balanced pool. At $k = 2$, Qwen3-8B answer accuracy rises from 43.0% to 53.7%, 54.8% and 52.6% across the three seeds (Table 1).

The 1–10 importance rating is the strongest untrained admission policy for Qwen3-8B on this pool. We evaluate it for all six models at all three budgets, and SDA exceeds it in 52 of 54 model–seed–budget comparisons. The two exceptions are Qwen3-14B at three stored concepts, where two of three seeds reach 44.9% against the rating’s 45.1%.

Qwen3-8B SDA reaches 40.5%, 53.7% and 51.8% answer accuracy at $k = 1, 2, 3$ on the statedness-balanced pool (Table 2). At $k = 2$, it exceeds the 1–10 importance rating by 5.2 points, implied-content wording by 9.1, the statedness-adjusted report by 14.6 and the combined teacher used directly by 8.8 (Table 8); it exceeds these baselines and output likelihood used directly at every budget. Section 6.4 separates the contributions of the teacher’s two signals.

Published admission criteria also fall below the untrained importance report on the primary pool. We implemented ConsistencyGate (Zhang and Li, 2026), A-MAC utility (Zhang et al., 2026), and Nemori’s prediction-error criterion (Ma et al., 2026) from their papers. Each admits top- k candidates under the same budget protocol (Table 2). At two stored concepts, SDA exceeds each Con-

Table 1: Answer accuracy (%) at $k = 2$ across models on the statedness-balanced pool (7.93 candidates per episode). Rating: 1–10 importance rating (Park et al., 2023). Gold-concept selection ranks the labeled useful concept first.

Model	Importance report	1–10 rating	Workspace readout	SDA (3 seeds)	Gold-concept selection
Qwen3-8B	43.0	48.5	43.0	53.7 / 54.8 / 52.6	54.3
Qwen3-14B	35.3	40.0	36.6	44.9 / 44.1 / 43.2	46.6
Qwen3-32B	45.7	43.5	46.0	52.1 / 54.3 / 53.4	54.8
Qwen2.5-7B	23.4	26.4	24.2	35.3 / 30.6 / 34.2	31.7
OLMo-2-7B	40.5	47.7	45.2	54.8 / 54.5 / 54.8	54.8
Mistral-7B	12.7	16.5	16.2	19.3 / 20.4 / 20.1	16.8

Table 2: Admission policies on the statedness-balanced pool, Qwen3-8B. Answer accuracy (%); ConsistencyGate includes sampled and single-pass forms, and prediction error is candidate surprisal under the passage. App. B.5 details published criteria.

Admission policy	$k = 1$	$k = 2$	$k = 3$
<i>Importance judgments of the untrained model</i>			
Importance report	30.6	43.0	45.5
1–10 importance rating (Park et al., 2023)	35.8	48.5	50.7
Implied-content wording	31.4	44.6	47.7
<i>Admission criteria from published methods</i>			
ConsistencyGate, sampled (Zhang and Li, 2026)	24.0	35.0	40.8
ConsistencyGate, single pass (Zhang and Li, 2026)	24.5	35.0	42.4
A-MAC utility (Zhang et al., 2026)	30.0	39.7	44.9
Prediction error (Ma et al., 2026)	16.5	25.9	30.3
<i>Statedness rules</i>			
Statedness-adjusted report	36.6	39.1	39.9
Reversed importance report	23.7	29.8	33.1
Unstated-first	28.6	32.0	36.4
Unstated-first with report ordering	36.9	39.1	39.9
<i>Teacher signals used directly</i>			
Output likelihood used directly	34.7	41.9	41.6
Combined teacher used directly	36.9	44.9	46.3
SDA	40.5 ± 0.3	53.7 ± 1.1	51.8 ± 0.5
Gold-concept selection	47.4	54.3	51.5

ConsistencyGate form by 18.7 points, A-MAC utility by 14.1 and prediction error by 27.8. The smallest margin is 6.9 points over A-MAC utility at three stored concepts (Table 8).

ConsistencyGate and A-MAC utility track whether the passage states a candidate. On the statedness-balanced pool, their correlations with statedness are +0.90 for single-pass ConsistencyGate, +0.85 for sampled ConsistencyGate, and +0.75 for A-MAC utility. On the standard pool, ConsistencyGate’s 0.7 threshold admits 26.7% of useful concepts against 70.8% of all candidates. Its within-episode AUC is 0.133 for single-pass scoring and 0.190 for sampled scoring.

A single statedness offset for stated candidates, fitted on development episodes, places every unstated candidate ahead of every stated candidate, turning the report into unstated-first with report ordering. It raises development AUC from 0.304

to 0.978 while lowering statedness-balanced AUC from 0.683 to 0.575. Accuracy matches this rule at $k = 2, 3$; at $k = 1$, the offset and rule reach 36.6% and 36.9%, respectively. Unlike this offset, SDA’s ranking depends on the passage (§6.4, Figure 4).

SDA improves retention of unstated useful concepts and also retains stated useful concepts. For Qwen3-8B at two stored concepts, the importance report retains the useful concept in 46.4% of stated-concept episodes and 33.0% of unstated-concept episodes. The workspace readout reaches 92.8% and 35.7%, respectively. Across the three seeds, SDA reaches 76.8%, 81.8% and 72.4% in stated-concept episodes and 64.8%, 64.8% and 64.3% in unstated-concept episodes.

The 173 episodes of the unstated-concept condition have no case-insensitive stem match for the useful concept in their original passages; all occupy odd positions and contain no added naming

Table 3: Closed-book and full-context readers, and admission from the context-only candidate extractor’s candidates, on the confirmation set, Qwen3-8B. Answer accuracy (%); admission stores two concepts. The extractor supplies 8.0 candidates per episode and includes the useful concept in 1.4% of episodes.

Condition	Accuracy
<i>Reader references</i>	
Closed-book reader	22.9
Full-context reader	63.9
<i>Admission from extracted candidates</i>	
Importance report	33.6
Gold-concept selection	34.4
SDA (3 seeds)	35.5 / 33.3 / 33.1

sentence. SDA reaches 37.4%, 50.1% and 46.6% answer accuracy at $k = 1, 2, 3$, compared with 24.9%, 40.5% and 37.0% for the importance report. The gains are 12.5, 9.6 and 9.6 points, respectively. SDA also exceeds the 1–10 importance rating by 5.6 points at two stored concepts.

On the standard pool, all 18 runs exceed the corresponding importance report, with a mean gain of 9.6 points (Tables 12 and 14). Both graders show these gains (App. D.1).

Replacing SDA’s retained useful concept with the next-ranked candidate lowers the frozen reader’s answer accuracy from 54.6% to 37.8%, a paired decrease of 16.9 points. SDA improves Qwen3-8B within-episode ranking and retention on the standard pool (Table 15). We replay SDA’s two selections on the 249 episodes where it retains the useful concept, then perform this replacement. The replacement loses 59 answers and gains 17.

Reader references and candidate extraction.

The closed-book and full-context readers reach 22.9% and 63.9% answer accuracy on Qwen3-8B, bracketing SDA’s 50.0% on the standard pool. With a context-only candidate extractor, SDA reaches 35.5%, 33.3% and 33.1% across the three seeds at $k = 2$ (Table 3; extraction prompt in App. B.6).

6.3 Robustness Across Candidate Pools

SDA’s answer accuracy gains extend to the naming-balanced pool, where naming carries no utility information, to the pool with unstated distractors, and to the enlarged pool. On the naming-balanced pool, SDA exceeds each model’s importance report in 17 of 18 model–budget cells across six models. The remaining cell differs by 0.1 points (Table 6). SDA exceeds the 1–10 importance rating

in all twelve model–budget cells at one and two stored concepts. At three stored concepts, SDA and the rating are within 2.6 points for every model. Qwen3-8B also improves within-episode ranking and useful-concept retention across seeds (Table 7).

On the pool with unstated distractors, SDA exceeds each model’s importance report in every model–seed–budget comparison (Table 10). For Qwen3-8B, SDA also exceeds output likelihood used directly at every budget on this pool and the enlarged pool (Tables 13 and 11).

At $k = 2$ on the pool with unstated distractors, SDA exceeds the importance report by 9.6 accuracy points, the 1–10 importance rating by 4.1, and unstated-first with report ordering by 5.8 (Table 14). On the enlarged pool, the corresponding gains are 11.0, 4.7, and 6.6 points (Table 14).

At one stored concept, SDA trails unstated-first with report ordering by 8.4 points on the pool with unstated distractors and by 5.5 points on the enlarged pool (Table 14). At two and three stored concepts, SDA leads every measured baseline on both pools (App. E.1).

6.4 Ablations

Teacher-signal ablations. Removing either teacher signal lowers answer accuracy on the paired-passage set. We remove either the workspace readout or output likelihood from SDA’s teacher and train the same adapter on the resulting labels, keeping training episodes and the step budget fixed.

Each candidate set appears twice under two passages with corresponding questions and different useful concepts. Both episodes include their partner’s useful concept, so only the passage identifies which concept is useful. We score 181 pairs on Qwen3-8B, Qwen3-14B, Qwen3-32B, Qwen2.5-7B, and OLMo-2-7B, yielding 1,810 observations per budget (Table 4; per-model results in Table 28).

At one stored concept, removing the workspace readout lowers answer accuracy by 1.9 points and retention by 7.3 points; removing output likelihood lowers answer accuracy by 1.3 points, and retention rises by 1.4 points. Both ablations lower mean accuracy, and removing the workspace readout also lowers retention, at every budget (Table 4). Differences are SDA minus the trained ablation, averaged over three seeds and pooled over models.

Passage replacement measures how the student’s selections depend on the passage (Figure 4). We score each episode before and after replacing its

Table 4: Paired-passage set results for Qwen3-8B, Qwen3-14B, Qwen3-32B, Qwen2.5-7B and OLMo-2-7B. Entries: SDA minus students trained with one teacher signal removed, pooled across models, in percentage points; retention indicates admission of the useful concept.

	Without the workspace readout		Without the output likelihood	
	Budget	Retention	Retention	Accuracy
$k = 1$		+7.3	+1.9	+1.3
$k = 2$		+4.6	+0.1	+1.6
$k = 3$		+3.0	+1.7	+1.4

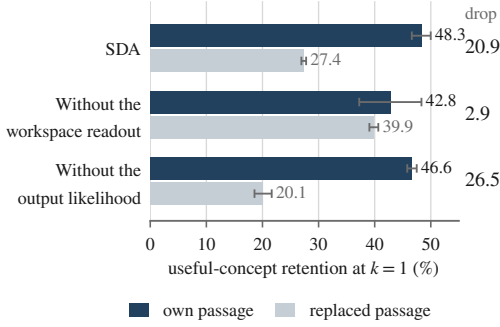


Figure 4: Passage replacement on the standard pool, Qwen3-8B. Retention is the percentage of episodes whose single stored concept is useful: own passage (solid), replaced passage (pale). Error bars: standard deviation over three seeds. Drops are own-passage minus replaced-passage retention; Table 30 includes the untrained workspace readout and importance report.

passage with another episode’s passage, keeping candidates, labels, and the question fixed. On 363 standard-pool episodes with one stored concept, Qwen3-8B SDA retention falls by 20.9 points. The student trained without the workspace readout loses 2.9 points; the student trained without output likelihood loses 26.5 points (Figure 4). Without the workspace readout, the student’s retention barely changes when the passage is replaced.

On the primary statedness-balanced pool, removing output likelihood lowers answer accuracy by 1.6 and 1.7 points at $k = 1, 2$; removing the workspace readout lowers it by 0.7, 0.6 and 0.3 points at $k = 1, 2, 3$ (Table 27). This analysis combines five models and 1,815 observations per budget, with per-model answer accuracy in Table 26.

Distillation versus direct teacher scores. On the standard pool at $k = 2$, SDA reaches 50.0% answer accuracy, compared with 45.7% for the combined teacher used directly and 40.2–40.5% for the validation-selected log-odds objectives (Tables 32 and 33; App. H.2).

To separate training gains from prompt changes, we train and evaluate Qwen3-8B SDA through the *implied-content wording*, which permits inferences without saying whether the passage states the useful concept. This wording raises standard-pool report AUC from 0.321 to 0.462 and improves answer accuracy at every budget (Table 17). With this wording, SDA reaches 37.1%, 50.7% and 51.0% answer accuracy at $k = 1, 2, 3$. It exceeds the prompt baseline at every budget (Table 17). SDA trained with passage-only wording reaches 37.6%, 50.0% and 52.2% at $k = 1, 2, 3$ (Tables 13 and 14).

6.5 Deployment

In an off-the-shelf serving engine, SDA keeps its answer accuracy at 242 candidates per second on one A100-80GB (Table 5). The served Qwen3-8B adapter reaches 50.1% answer accuracy on the standard pool, within 0.3 percentage points of the reference selections, the primary SDA seed’s selections in the main evaluation. The two selections differ in answer correctness on five episodes, split two and three. Top-2 memories agree on 91.7% of episodes, and pooled AUC differs by at most 0.001.

The throughput measurement uses batched serving with prefix caching over 1,607 candidates after warm-up. SDA takes 4.1 ms per candidate; the importance report reaches 290 candidates/s at 3.4 ms per candidate (Table 5).

7 Implications

SDA raises pooled AUC on the development sets at every tested Qwen3 training scale and seed, changing the admission score while the adapter-enabled model’s workspace readout AUC changes by at most 0.0046 (App. G.4).

Separating usefulness from statedness in admission evaluation requires comparing policies on the same episodes with a fixed reader and budget, balancing the statedness of the useful concept, and including unstated distractors. Evaluation should

Table 5: Admission serving on Qwen3-8B, standard pool, one A100-80GB with prefix caching. Amortized time is per candidate; top-2 selections match the primary SDA seed’s main-evaluation selections on 91.7% of episodes, and pooled AUC differs by at most 0.001.

Admission scoring	Time (ms/candidate)	Candidates/s	Hidden states
Importance report	3.4	290	not needed
SDA adapter	4.1	242	not needed

report within-episode ranking, retention split by statedness, and answer accuracy (§4).

8 Conclusion

The importance report favors candidates stated in the passage, even when the useful concept is unstated. SDA trains a LoRA adapter through the model’s importance prompt on labels that the frozen model derives from its workspace readout and output likelihood, improving answer accuracy over each model’s importance report on the statedness-balanced pool. The adapter attaches only to admission requests and preserves answer accuracy in an off-the-shelf serving engine. SDA lets an agent decide what to retain before a question arrives and store the unstated concepts that later answers need.

9 Limitations

The evaluation covers admission of host-supplied candidates in generated episodes with one later question; end-to-end long-conversation memory also requires candidate extraction, update, retrieval, and deletion. Gold utility labels mark the generator’s useful concept, while the optimal memory can differ by reader. Stem matching approximates statedness, and semantic grading uses an LLM judge. Serving and full-candidate workspace readout measurements (which score every token of a candidate) cover Qwen3-8B on the standard pool; serving uses batched scoring on one A100-80GB. The objective ablation measures ranking and answer accuracy, with three seeds at learning rate 10^{-5} and one at each smaller learning rate. Because the 54 model–seed–budget comparisons share episodes, they measure consistency across settings, not 54 independent replications.

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A Set Construction and Leakage Audit

A.1 Passages, concepts and future questions

The primary generator is GPT-4.1. Each episode contains a passage, candidate concepts with gold utility labels, and a future question and answer. Explicit places the useful concept in the passage. **Implied-answer** training episodes cue an unstated useful concept that supplies the answer, including episodes from the alternate-generator set. Table 22 reports this construction as Implied-answer. Decoupled requires an unstated bridge to support a further inference: the answer is absent from the passage and candidate set, and the question shares no passage content word except a protagonist’s name. Compositional requires a multi-hop inference and permits question anchors from the passage. Decoupled and Compositional are development sets used for configuration selection and excluded from gradient training.

Rule-based validation checks candidate strings, bridge absence, answer exclusion and the Decoupled question’s lexical overlap. Concept deduplication removes repeated concepts across retained episodes, while the leakage audit also checks full lexemes, inflected answers and question anchors. A blind GPT-6-Astra alias audit receives the future question, gold answer and bridge. It rejects identity, morphological or derivational variants, synonyms, translations, abbreviations and containment that trivially yields the answer. For example, *caterer/catering* is an alias, whereas *Madrid/Spanish* requires a further inference.

A.2 Episode-generation prompt

GPT-4.1 receives the system instruction, “You are a careful dataset designer for a cognitive-science experiment on LLM memory.” The following prose transcription preserves the generation constraints (Prompt 1); the output-format instruction requests one structured record per episode. Generation uses temperature 1.0 and batches of up to five episodes, after which rule-based filters and the alias audit apply.

A.3 Held-out and replication sets

The interim held-out set and confirmation set use the Decoupled construction prompt and rule-based filters, followed by deduplication and disjointness filtering. A context match after lowercasing and trimming against a prior set causes exclusion. A bridge that matches a labeled useful concept of the

training episodes also causes exclusion. The confirmation set also excludes contexts appearing in the interim held-out set. Shared answer categories, such as languages and currencies, are permitted. The retained sets contain 316 and 363 episodes, respectively. The alternate-generator set repeats the Implied-answer construction with GPT-4o, and the entity-bridge set comes from GPT-5.6 (§3).

A.4 Reconstructing the reported controls

All pool interventions traverse the 363 confirmation-set episodes in their fixed order and preserve the question, answer and original utility labels. Added candidates receive negative utility labels. Within an episode, original candidates precede additions, whose recorded order is retained. The archived control file records the passages, ordered additions, named candidates and passage permutation; replay uses these recorded draws. The standard pool contains 1,607 candidates, 4.43 per episode. Statedness uses the case-insensitive stem match specified in §4.

Worked source episodes. Figure 5 gives the first two confirmation-set episodes in full; the second is the episode drawn in Figure 2.

Stated-bridge, position and distractor-named controls. The exact sentence form is “The passage also mentions *concept*.” The stated-bridge control appends this sentence with the useful concept, whereas the position control prepends it. The distractor-named control appends it with the first non-useful candidate in the original order. Thus Yara’s stated-bridge control adds “The passage also mentions insurance.” after the passage, the position control puts that sentence before “Yara compared,” and the distractor-named control instead appends “The passage also mentions insurer.”

Pool with unstated distractors. An LLM proposes four single-word concepts from the useful concept’s semantic category, asking for alternatives a careless reader might confuse with it but that the passage does not imply. The instruction requires lowercase words absent from the passage and a comma-separated response. We retain the response order, discard a suggested concept that matches an original candidate or occurs as a case-insensitive substring of the passage, and append the accepted concepts. The pool contains 1,450 additions: four in 361 episodes and three in episodes 24 and 263 after filtering, giving 3,057 candidates

Example 1: First confirmation-set episode	Example 2: Second confirmation-set episode
Passage. Yara compared renewal quotes from different insurers online and reviewed her accident history. She called the hotline to ask clarifying questions about premium increases. After considering the coverage limits and deductibles, Yara finally entered her payment details to confirm the policy. Candidates, in supplied order. insurer, deductible, history, policy, insurance. Useful concept. insurance. Surface distractors. deductible, history. Neutral concepts. insurer, policy. Future question. Which professional would be most relevant for Yara to contact regarding her recent purchase? Answer. agent.	Passage. Jibril had always wanted to gaze up at the world’s highest outdoor observation deck. Early one morning, he took a swift elevator to level 148 of the Burj Khalifa, joining a crowd of eager tourists. Below him, the city sparkled with the first rays of the desert sun. After snapping several panoramic photos, he browsed the sleek gift shop for souvenirs. Candidates, in supplied order. elevator, desert, gift, observation, dubai, tourists. Useful concept. dubai. Surface distractors. desert, tourists. Neutral concepts. elevator, gift, observation. Future question. What language was Jibril most likely to hear from local staff during his visit? Answer. arabic.

Figure 5: First two confirmation-set episodes in full. Admission sees only the passage and candidate strings; utility labels, questions and answers are included for explanation.

in total (8.42 per episode). For Yara, the ordered additions are *warranty*, *mortgage*, *pension*, *annuity*. The development confusable-distractor control applies the same procedure to Decoupled, whereas the unrelated-distractor control takes bridges from subsequent episodes, as described for the enlarged pool below.

Statedness-balanced pool. Odd positions, counting from one, keep the original passage; even positions append “The passage also mentions *concept*.” with the useful concept substituted. Every episode keeps its original candidates and takes the first four accepted additions at odd positions or the first three at even positions from the pool with unstated distractors, preserving their order. This yields 2,877 candidates, 7.93 per episode. Yara occupies position one, so her passage is unchanged and her pool is the five original candidates followed by *warranty*, *mortgage*, *pension*, *annuity*. Jibril occupies position two, so “The passage also mentions dubai.” is appended to Example 2’s passage and *doha*, *riyadh*, *muscat* are appended to its six candidates.

Naming-balanced pool. This pool uses the same ordered candidate lists as the statedness-balanced pool but starts from the original passages and names four candidates per episode. At odd positions, the useful concept is excluded and four other candidates are sampled uniformly without replacement; at even positions, the useful concept is included and three others are sampled. One Python standard pseudorandom generator seeded with zero processes episodes in fixed order. The useful con-

cept is prepended to the sampled list when included, and each selected list is shuffled before sentences are constructed. Sampling and shuffling consume the same continuing random stream. The five sentence forms, in rotation, are “The passage also mentions *concept*.”, “*concept* is relevant here.”, “Note that *concept* comes up as well.”, “There is also *concept* to consider.”, and “*concept* features in this as well.”

Episodes and insertions are numbered from zero for the placement rule. The first selected candidate uses the sentence form at the episode’s position in the five-form cycle; each subsequent candidate advances one form. Trailing full stops are stripped from the original passage, which is split at full-stop-space boundaries; each segment is trimmed and one full stop restored to it. Before each insertion, the current sentence list is counted, including earlier additions. With episode index e , insertion index j and current sentence count n , the insertion boundary is $(e + 3j) \bmod (n + 1)$. Boundary zero precedes the passage; the final boundary follows it.

For Yara, the shuffled selection is *pension*, *annuity*, *insurer*, *policy*. The additions are “The passage also mentions pension.”, “annuity is relevant here.”, “Note that insurer comes up as well.”, and “There is also policy to consider.” Their successive insertion boundaries are zero, three, zero and two. The final passage begins with the insurer, pension and policy sentences, then Yara’s first two original sentences, then the annuity sentence, then her final original sentence. Across 2,877 candidates, there are 1,452

naming sentences, 181 of which name the useful concept.

Enlarged pool. This pool combines the standard candidate list, the accepted same-topic additions, and four bridges from other episodes. To choose the latter, the procedure starts with the next episode in fixed order, advances cyclically, and accepts a bridge only if it is absent from the passage, absent from the original candidate list and distinct from earlier accepted bridges. Selection stops after four accepted bridges. The three lists are concatenated in this order, retaining the first occurrence of each concept. The resulting 4,506 candidates average 12.41 per episode. Yara’s pool is *insurer, deductible, history, policy, insurance, warranty, mortgage, pension, annuity, dubai, church, kanji, doctor*. The unrelated-distractor control uses just the standard candidates and the four cross-episode additions; for Yara these are *dubai, church, kanji, doctor*.

Passage replacement and bridge replacement. Passage replacement uses the archived episode permutation, replacing each passage while preserving its candidates, utility labels, question and answer. The first episode receives the passage at position 105, about Lars skating on slick tracks with thin-bladed boots; its candidates remain Yara’s five concepts and its answer remains *agent*. The bridge-replacement control acts on SDA’s recorded top-two selections only when they contain the useful concept: the other selection is retained and the useful concept is replaced with the next-ranked candidate, then the same frozen reader answers the same question again. For a selection of *dubai, elevator* with *gift* next, the replayed memory is *gift, elevator*. The 249 measured replays apply this rule to the recorded selections (§E.2).

Candidate extraction and reader references. The context-only candidate extractor receives the original passage and the instruction in Appendix B.6; the future question and bridge are hidden. Its parsed output replaces the supplied candidate list in extraction order. For Yara the resulting candidates are *online, accident, hotline, premium, coverage, deductibles, payment, policy*, so *insurance* is unavailable to admission. At two stored concepts, the same frozen reader answers from each policy’s selected pair. The closed-book reference gives that reader Yara’s question with no memory, whereas the full-context reference gives

the same question and her original three-sentence passage. Neither reference performs admission.

Wording and replication controls. The wording controls change only the admission instruction, whose five exact forms appear in Appendix F.3. The alternate-generator and entity-bridge sets repeat the episode construction with the generators and concept categories of App. G.3.

Naming-balanced ranking and retention. SDA improves ranking and retention under the naming-balanced construction (Table 7). Across Qwen3-8B seeds, within-episode AUC is 0.715–0.725, versus 0.566 for the report and 0.636 for the workspace readout. Useful-concept retention at $k = 2$ is 53.2–54.0%, versus 24.0% and 40.8%, respectively.

B SDA Training and Scoring Details

B.1 Teacher construction and data split

Within each episode, let $r_S(c)$ be a candidate’s ascending mid-rank under score S , with ties assigned the average of their occupied ranks. For $m > 1$ candidates, the percentile is

$$P_S(c; x) = \frac{r_S(c) - 1}{m - 1}, \quad S \in \{W, LL\}. \quad (7)$$

For a singleton pool the percentile is one half. Averaging the percentiles yields the teacher in Eq. (4); descending teacher score and then candidate order determine the top-2 labels.

A source group is one generator-and-construction group: Explicit (88 episodes), implied-answer from the primary generator (75), or implied-answer from the alternate generator (57). The split is stratified by these groups and assigns whole episodes, including all their candidates, to training or validation. Within each group, episodes are sorted by a cryptographic hash of the fixed split seed and the source-qualified episode identity; the lowest 20%, rounded up, enter validation. The split seed is zero and stays fixed across optimization seeds. The three groups respectively contain 70 training and 18 validation episodes, 60 training and 15 validation episodes, and 45 training and 12 validation episodes, giving 175 training and 45 validation episodes in total. The selection protocol is specified in §5.2.

B.2 Complete training configuration

LoRA adapts the attention query, key, value and output projections and the feed-forward gate, up

Table 6: Answer accuracy (%) on the naming-balanced pool, six models, at one, two and three stored concepts. Report: each model’s importance report; Rating: 1–10 importance rating (Park et al., 2023).

Model	$k = 1$			$k = 2$			$k = 3$		
	Report	Rating	SDA	Report	Rating	SDA	Report	Rating	SDA
Qwen3-8B	27.6	35.3	36.4 ± 0.5	42.7	45.2	46.2 ± 0.2	46.3	47.1	47.0 ± 0.2
Qwen3-14B	21.8	30.3	33.7 ± 0.4	33.3	36.4	40.8 ± 0.6	41.3	43.2	41.2 ± 0.4
Qwen3-32B	31.7	30.0	41.1 ± 0.6	43.2	43.0	48.9 ± 1.0	50.7	51.8	53.4 ± 1.0
Qwen2.5-7B	9.4	11.6	13.3 ± 1.3	21.2	25.6	28.3 ± 0.6	28.6	33.3	33.0 ± 0.2
OLMo-2-7B	24.8	30.6	37.0 ± 0.3	39.4	43.0	46.7 ± 1.1	44.6	52.1	49.5 ± 0.8
Mistral-7B	4.7	5.8	9.9 ± 1.1	9.6	13.2	16.0 ± 0.7	19.8	21.5	23.7 ± 1.7

Table 7: Candidate ranking and useful-concept retention on the naming-balanced pool, Qwen3-8B, 2,877 candidates. Entries are AUC and retention (%).

Admission policy	Pooled	Within-episode			
	AUC	AUC	B_1	B_2	B_3
Importance report	0.570	0.566	9.4	24.0	42.7
Workspace readout	0.635	0.636	22.0	40.8	54.8
SDA, seed 0	0.726	0.725	36.1	54.0	67.5
SDA, seed 1	0.714	0.715	32.2	54.0	65.6
SDA, seed 2	0.728	0.725	35.5	53.2	68.0

and down projections. Base parameters remain frozen. Only the target word contributes to the vocabulary-normalized cross-entropy in Eq. (6). Prompt tokens are masked. Qwen3 uses its chat template with thinking disabled.

Setting	Configuration
Adapter	Low-rank adaptation, rank 16, scaling 32; attention and feed-forward projections
Optimizer	AdamW; constant learning rate 1×10^{-5}
Batching	Batch size 1; gradient accumulation over 4 batches
Training duration	Two epochs; 414 optimizer steps
Sequence length	At most 1024 tokens
Data	175 training and 45 validation episodes; source-stratified, disjoint episodes
Targets	Two highest teacher ranks positive; all others negative

B.3 Measurement conventions

The importance report in every main table is scored in log space over the yes and no token sets. The wording comparison uses a common scorer for all five wordings, so its rows are comparable under that aggregation. Mistral-7B’s passage-only wording gives AUC 0.411 in the main tables and 0.440 in the wording comparison because the two scorers aggregate differently. Scoring the same wording one prompt at a time also gives 0.440, identical to the batched result. On Qwen3-8B’s standard pool, first-token workspace AUC is 0.543 at the end of the passage. Appending the candidate and reading the state immediately before its first token

gives 0.530. The latter is the matched baseline for full-candidate AUC 0.574. The untrained report’s answer accuracy is 40.8% in the main tables and 40.5% in the objective ablation, which evaluates the untrained checkpoint through the adapter-loading path instead of the frozen path.

B.4 Admission baselines

SDA is compared with the following baselines. The **importance report** uses the untrained model’s score V in Eq. (5). The **1–10 importance rating** is the expected rating under normalized probabilities of the complete answer strings. The **reversed importance report** admits the lowest-scoring candidates. **Unstated-first** prioritizes candidates absent from the passage under the statedness test and breaks ties at random. **Unstated-first with report ordering** prioritizes unstated candidates and uses the importance report to order candidates within each statedness class. The **statedness-adjusted report** adds a single coefficient times the candidate’s statedness indicator to the untrained report, with the coefficient fitted on development episodes and held fixed for evaluation. The **combined teacher used directly** ranks by Eq. (4) without training an adapter. The **implied-content wording** permits implied content; the **statedness-cued wording** explicitly says the passage may never mention the useful concept (App. F.3). **Random selection** samples candidates uniformly; **output likelihood** and the **workspace readout** rank candidates directly

by their respective scores. **Gold-concept selection** ranks the labeled useful concept first; the closed-book and full-context readers provide accuracy references.

B.5 Published admission criteria as implemented

ConsistencyGate. ConsistencyGate (Zhang and Li, 2026) is a write-time gate that scores how strongly the source context supports a candidate fact. We measure two forms. The sampled form follows their Equations 2 and 3, using their Appendix C prompt verbatim with five samples at temperature one and at most eight new tokens per sample. Its score is the mean of the parsed decimals. The single-pass form follows their Equation 4 variant: the same question is posed as a yes/no question, and the score is the probability of yes from a softmax over the yes and no token log-probabilities. Ranking uses the score itself under the shared top- k budget, and their 0.7 threshold enters only the gate-rate diagnostic.

A-MAC. A-MAC (Zhang et al., 2026) is a multi-factor value model. Utility is obtained from one temperature-zero call asking whether the candidate is actionable, supports likely follow-up questions, or captures a persistent constraint or preference. Confidence is the maximum ROUGE-L between the candidate and a supporting passage sentence. The type prior is their rule-based entity cue. Their novelty term compares against memory already held, and their recency term decays with elapsed time. Within an episode, memory is empty at the first admission and all candidates share one timestamp, so neither term orders candidates. These two terms are therefore omitted from the ranking score. The three remaining factors are combined as a weighted sum, with the utility weight fixed at one. Confidence and type-prior weights are searched on a grid from 0.0 to 2.0 in steps of 0.1, selected by within-episode AUC inside five-fold cross-validation over the 363 episodes, and evaluated out of fold. Every fold selects zero weight for both confidence and type prior, so the fitted combination coincides with utility alone. Out-of-fold within-episode AUC is 0.6795 for both.

Nemori. Nemori’s prediction-error criterion (Ma et al., 2026) keeps what the model fails to predict. Applied to a candidate concept and its passage, the score is the candidate’s surprisal under the passage, the negative of the output likelihood. The criterion

ranks candidates for admission, the part of Nemori that applies to this setting; memories already held play no role.

All three score the same unit as every other policy, a candidate concept and its episode passage with the future question hidden, and admit the top- k under the same budget protocol. Prompts 4 to 6 give the three scoring instructions.

B.6 Importance prompt, reader, rating and extraction scoring

Appendix I lists the prompts. Each is a user message rendered with the model’s chat template and assistant generation prefix; Qwen3 thinking is disabled. SDA and the importance report share the importance prompt (Prompt 2).

The affirmative token set contains the distinct first tokens of *yes* in lowercase, initial-capital and uppercase forms, with and without a leading space. The negative set is constructed analogously from *no*. Equation (5) gives the common report score; SDA applies the same scoring rule with its adapter enabled.

The reader (Prompt 8) uses the frozen model with its admission adapter disabled. The closed-book reference asks “Answer this question concisely.” followed by the question and answer prefix. The full-context reference supplies the passage, then “Answer concisely.” and the same question and answer prefix. Greedy decoding is used throughout.

Let a_r be the complete answer string for rating r , and let $u_R(c, x)$ be the rating prompt (Prompt 3). Each answer receives its full sequence probability, multiplying the conditional probabilities of its tokens. The score is

$$R(c; x) = \sum_{r=1}^{10} r \pi_r(c; x), \quad (8)$$

$$\pi_r(c; x) = \frac{p_{\mathcal{M}}(a_r \mid u_R(c, x))}{\sum_{j=1}^{10} p_{\mathcal{M}}(a_j \mid u_R(c, x))}.$$

A rating represented by multiple tokens is scored as a complete answer.

The context-only candidate extractor (Prompt 7) uses Qwen3-8B with greedy decoding. Distinct parsed concepts form the candidate pool. The useful concept, future question and answer are withheld from extraction and carried forward for evaluation. Useful-concept coverage accepts case differences and a trailing plural ending.

B.7 Ablation protocol

The teacher and optimization ablations in §6.4 use the same training episodes, adapter shape and matched step budget. The accuracy-reward condition samples admitted sets from the admission scores and asks the frozen reader to answer from the selected memory. Answer correctness supplies the reward. Within each rollout group, reward differences are normalized into advantages; updates use a clipped group-relative policy objective and a penalty for departure from the reference policy (Shao et al., 2024). The future question enters the reward computation. The refinement condition applies these updates after SDA training.

C Held-Out Set Locking

The interim held-out set and the confirmation set were locked on 15 September 2026. Each set’s episode count and cryptographic digest were recorded before evaluated models read it: 316 episodes for the interim held-out set and 363 for the confirmation set. The interim held-out set supplied the initial held-out evaluation, while the confirmation set was locked after configuration selection. Its primary endpoint compares SDA with the importance report; a primary training seed was designated and the remaining seeds reported alongside it. Its secondary endpoint compares SDA with the rating. The confirmation set uses the blind semantic judge in App. D.1.

D Answer Grading and Serving Protocol

D.1 Blind semantic judge

The judge, GPT-6-Astra, receives the question, gold answer and reader response with no admission-policy identity. Prompt 9 quotes the substantive grading instruction verbatim; its output-format instruction requests a label for every supplied answer. Only correct responses contribute to answer accuracy; hedged and incorrect responses count as errors. Judgments are reused for identical question, gold answer and response triples. Each answer is also graded by string matching after normalization. Agreement is 85.1% across 13,068 answers; every training run exceeds its own report under both graders.

D.2 Amortized scoring time and throughput

The timing evaluation loads Qwen3-8B on one A100-80GB and constructs an importance prompt for each of the 1,607 standard-pool candidates.

vLLM enables prefix caching and LoRA serving, warms the engine on an initial subset, and times batched generation over the full candidate list. Each prompt generates one token greedily; SDA attaches the adapter to the admission request, while the importance report uses the same engine without it. Model loading is outside the timed region. Milliseconds per candidate are wall time divided by candidate count; throughput is candidate count divided by wall time. Reference selections are the primary SDA seed’s selections in the main evaluation; both served and reference selections are passed to the same frozen reader and semantic judge. The serving engine recovers yes/no masses from returned token log probabilities.

D.3 Capability with the adapter enabled on every token

Enabling the adapter on every token changes the four measured capability scores by at most 0.3 percentage points under matched prompts. Base and adapter scores are 71.6% and 71.5% on MMLU (Hendrycks et al., 2021), 80.4% and 80.7% on ARC-Easy, 56.2% and 56.4% on ARC-Challenge (Clark et al., 2018), and 88.4% and 88.6% on the GSM8K evaluation subset (Cobbe et al., 2021).

E Answer Accuracy across Evaluation Sets and Pools

Table 8 gives the paired differences behind §6.2.

Qwen3-8B SDA improves mean answer accuracy over the report on each development set and held-out set in Table 9. The gains also occur with unstated distractors and in the enlarged pool (Tables 10 and 11). Across all six models, mean SDA answer accuracy exceeds the report and rating at every standard-pool budget (Tables 12 and 13).

E.1 Candidate-pool comparisons

Standard pool. Because the useful concept is usually the sole unstated candidate on this pool (§6.1, App. F.2), unstated-first reaches 46.6%, 53.4% and 49.6% answer accuracy at $k = 1, 2, 3$; unstated-first with report ordering reaches 47.1%, 53.4% and 52.6%, and the reversed importance report reaches 35.5%, 46.6% and 49.0%. SDA’s corresponding means are 37.6%, 50.0% and 52.2%.

Pool with unstated distractors. On the pool with unstated distractors, SDA exceeds each model’s importance report under blind semantic grading in 54 of 54 comparisons across six models,

Table 8: Paired SDA minus baseline answer-accuracy differences (percentage points) for each baseline in Table 2, statedness-balanced pool, Qwen3-8B. Published methods: 1–10 importance rating (Park et al., 2023); Consistency-Gate (Zhang and Li, 2026); A-MAC (Zhang et al., 2026); prediction error (Ma et al., 2026).

Baseline	$k = 1$	$k = 2$	$k = 3$
<i>Importance judgments of the untrained model</i>			
Importance report	+9.9	+10.7	+6.3
1–10 importance rating	+4.7	+5.2	+1.1
Implied-content wording	+9.1	+9.1	+4.1
<i>Admission criteria from published methods</i>			
ConsistencyGate, sampled	+16.5	+18.7	+11.0
ConsistencyGate, single pass	+16.0	+18.7	+9.4
A-MAC utility	+10.5	+14.1	+6.9
Prediction error	+24.0	+27.8	+21.5
<i>Statedness rules</i>			
Statedness-adjusted report	+3.9	+14.6	+11.8
Reversed importance report	+16.8	+24.0	+18.7
Unstated-first	+11.8	+21.8	+15.4
Unstated-first with report ordering	+3.6	+14.6	+11.8
<i>Teacher signals used directly</i>			
Output likelihood used directly	+5.8	+11.8	+10.2
Combined teacher used directly	+3.6	+8.8	+5.5

Table 9: Answer accuracy (%) at $k=2$, Qwen3-8B: development sets (left) and the two held-out sets built with the Decoupled construction (right). Rating: 1–10 importance rating (Park et al., 2023).

Admission policy	Development				Held out	
	Decoupled $n=68$	Compositional $n=52$	Confusable $n=68$	Unrelated $n=68$	Interim held-out $n=316$	Confirmation $n=363$
Importance report	35.3	46.2	35.3	36.8	34.8	40.8
1–10 importance rating	47.1	53.8	48.5	48.5	43.7	45.7
SDA	49.5 ± 2.2	54.5 ± 1.1	48.5 ± 2.5	50.0 ± 2.5	44.2 ± 0.7	50.0 ± 1.4
Gold-concept selection	51.5	65.4	51.5	51.5	50.0	54.3

Table 10: Answer accuracy (%) across models on the pool with unstated distractors (confirmation set, 8.4 candidates per episode) at $k = 2$, with SDA values across the three seeds.

Model	Importance report	SDA (3 seeds)	Gold-concept selection
Qwen3-8B	41.0	51.0 / 51.2 / 49.9	54.3
Qwen3-14B	32.2	40.8 / 39.4 / 39.9	46.6
Qwen3-32B	41.0	51.0 / 50.7 / 50.1	54.8
Qwen2.5-7B	23.7	31.4 / 28.4 / 30.6	31.7
OLMo-2-7B	38.8	52.3 / 51.0 / 51.2	54.8
Mistral-7B	12.9	19.0 / 19.6 / 17.9	16.8

three seeds and budgets $k = 1, 2, 3$ (Table 10). On Qwen3-8B, output likelihood used directly reaches 30.6%, 39.1% and 40.5% answer accuracy at $k = 1, 2, 3$, against SDA’s means of 37.6%, 50.7% and 51.6%. SDA leads the measured baselines at $k = 2, 3$; unstated-first with report ordering leads at $k = 1$ (Tables 13 and 14). At $k = 2$, unstated-first falls from 53.4% to 35.3% and the reversed importance report from 46.6% to 28.9%, both below the importance report’s 41.0%; SDA reaches 51.0%, 51.2% and 49.9% across the

three seeds. Every SDA seed exceeds unstated-first and the reversed importance report at every budget. SDA reaches 37.7%, 36.4% and 38.6% across the three seeds at $k = 1$ and 51.8%, 51.5% and 51.5% at $k = 3$; the rating reaches 35.8% and 48.8%.

Stated-bridge control and entity-bridge set. On the stated-bridge control at $k = 2$, SDA reaches 55.4%, 55.6% and 52.9% answer accuracy across the three seeds, against 44.9% for the report and 39.9% for the reversed importance report. On the

Table 11: Answer accuracy (%) on the enlarged pool (confirmation set, 12.41 candidates per episode), Qwen3-8B.

Budget	Importance report	Output likelihood	Workspace readout	SDA (3 seeds)	Gold-concept selection
$k = 1$	29.5	28.1	24.0	38.0 / 38.0 / 39.7	47.1
$k = 2$	40.5	32.0	34.7	50.7 / 52.6 / 51.2	54.3
$k = 3$	41.9	33.9	38.6	52.9 / 52.1 / 52.1	51.5

Table 12: Answer accuracy (%) across models on the standard pool of the confirmation set, with k stored concepts. Rating: 1–10 importance rating (Park et al., 2023).

Model	$k=1$			$k=2$			$k=3$		
	Report	Rating	SDA	Report	Rating	SDA	Report	Rating	SDA
Qwen3-8B	28.1	34.7	37.6 $\pm$ 1.0	40.8	45.7	50.0 $\pm$ 1.4	41.6	47.9	52.2 $\pm$ 1.1
Qwen3-14B	23.4	31.1	33.5 $\pm$ 1.4	32.5	39.1	40.5 $\pm$ 1.4	37.5	43.5	43.6 $\pm$ 0.8
Qwen3-32B	32.5	32.0	41.4 $\pm$ 0.3	41.0	44.6	50.2 $\pm$ 0.8	46.0	50.7	53.0 $\pm$ 0.3
Qwen2.5-7B	9.6	9.6	16.8 $\pm$ 2.9	23.1	22.6	29.8 $\pm$ 2.1	29.5	35.0	35.8 $\pm$ 0.8
OLMo-2-7B	27.5	33.1	48.5 $\pm$ 0.6	38.0	44.1	55.4 $\pm$ 0.3	44.4	52.6	56.9 $\pm$ 0.6
Mistral-7B	5.2	9.6	12.4 $\pm$ 1.1	12.7	15.7	19.8 $\pm$ 0.8	19.3	22.0	24.2 $\pm$ 1.1

Table 13: Admission policies on the standard pool and pool with unstated distractors, Qwen3-8B. Answer accuracy is in percent; rows above the rule are untrained baselines, and dashes denote unmeasured settings.

Admission policy	Standard pool			Pool with unstated distractors		
	$k = 1$	$k = 2$	$k = 3$	$k = 1$	$k = 2$	$k = 3$
Importance report	28.1	40.8	41.6	28.6	41.0	41.3
1–10 importance rating (Park et al., 2023)	34.7	45.7	47.9	35.8	46.6	48.8
Reversed importance report	35.5	46.6	49.0	23.4	28.9	30.9
Unstated-first	46.6	53.4	49.6	31.7	35.3	36.9
Random selection	28.6	44.6	46.6	—	—	—
Output likelihood	—	48.8	—	30.6	39.1	40.5
Workspace readout	29.8	44.6	49.9	26.7	39.4	44.9
SDA	37.6 $\pm$ 1.0	50.0 $\pm$ 1.4	52.2 $\pm$ 1.1	37.6 $\pm$ 1.1	50.7 $\pm$ 0.7	51.6 $\pm$ 0.2
Gold-concept selection	47.1	54.3	51.5	47.1	54.3	51.5

entity-bridge set, the reversed importance report reaches pooled AUC 0.281, while SDA reaches 0.987, 0.980 and 0.987 across the three seeds, with within-episode AUC 1.000. Reversing the report helps only when it prefers stated candidates while the useful concept is unstated (§6.1). On the entity-bridge set, the reports of Qwen2.5-7B and Qwen3-8B prefer unstated candidates (App. G.3), so reversal ranks the useful concept low.

Enlarged pool. Across the three seeds, SDA reaches 38.0%, 38.0% and 39.7% at $k = 1$, 50.7%, 52.6% and 51.2% at $k = 2$, and 52.9%, 52.1% and 52.1% at $k = 3$ (Table 11). The rating reaches 36.4%, 46.8% and 48.2%, respectively; every SDA seed leads it at every budget. SDA exceeds the rating by 2.2, 4.7 and 4.1 points at $k = 1, 2, 3$ in paired mean answer accuracy (Table 14). Unstated-first with report ordering has the

highest answer accuracy at $k = 1$; SDA leads the measured baselines at $k = 2, 3$. Output likelihood used directly reaches 28.1%, 32.0% and 33.9% at $k = 1, 2, 3$, against SDA’s means of 38.6%, 51.5% and 52.4%. Random selection reaches 21.5%, 27.0% and 29.8%, the report 29.5%, 40.5% and 41.9%, and the workspace readout 24.0%, 34.7% and 38.6% at $k = 1, 2, 3$.

E.2 Within-episode ranking and useful-concept retention

SDA improves both within-episode ranking and retention of the useful concept at $k = 2$ (Table 15). Conditional accuracy separates retaining the useful concept from answering the question: for the primary SDA seed, accuracy is 54.6% when the useful concept is retained and 41.2% when it is not, versus 48.2% and 37.7% for the report.

Table 14: Paired SDA minus baseline answer-accuracy differences on three candidate pools, Qwen3-8B. Entries are in percentage points.

Baseline	$k = 1$	$k = 2$	$k = 3$
<i>Standard pool</i>			
Importance report	+9.6	+9.3	+10.6
1–10 importance rating	+2.9	+4.3	+4.2
Statedness-cued wording	+5.4	+7.1	+6.2
Reversed importance report	+2.1	+3.5	+3.1
Unstated-first	−8.9	−3.4	+2.6
Unstated-first with report ordering	−9.5	−3.4	−0.5
Random selection	+9.0	+5.4	+5.6
<i>Pool with unstated distractors</i>			
Importance report	+8.9	+9.6	+10.3
1–10 importance rating	+1.7	+4.1	+2.9
Reversed importance report	+14.1	+21.8	+20.8
Unstated-first	+5.9	+15.4	+14.7
Unstated-first with report ordering	−8.4	+5.8	+8.9
Output likelihood	+7.0	+11.6	+11.1
<i>Enlarged pool</i>			
Importance report	+9.1	+11.0	+10.5
1–10 importance rating	+2.2	+4.7	+4.1
Unstated-first with report ordering	−5.5	+6.6	+12.1
Output likelihood	+10.5	+19.6	+18.5
Random selection	+17.1	+24.5	+22.6

The bridge-replacement test replays the primary SDA seed’s two selections on the 249 episodes whose memory contains the useful concept. We replace only that concept with SDA’s next-ranked candidate, keeping the other selection, question and frozen reader fixed; §6.2 reports the results.

Gold-concept selection and answer accuracy. Qwen3-8B reaches 54.3% answer accuracy with gold-concept selection and 50.0% with SDA. The reader answers from the complete stored set and its own knowledge. On OLMo-2-7B and Mistral-7B, SDA reaches 55.4% and 19.8%, versus 54.8% and 16.8% for gold-concept selection.

F Statedness, Wording and Pool-Composition Measurements

F.1 Ranking among candidates with the same statedness

Restricting the pool with unstated distractors to episodes with an unstated useful concept and at least one unstated negative leaves 345 episodes, with 5.01 qualifying candidates per episode on average. §6.1 reports the resulting AUCs (Table 16). This measurement holds statedness fixed; the full-pool comparison measures the competition between stated and unstated candidates.

F.2 Pooled AUC and comparison-pair composition

On the confirmation set, the useful concept is stated in 5% of episodes and negative candidates in 99%; in 95% of episodes, the useful concept is the sole unstated candidate. Adding unstated distractors leaves every original Qwen3-8B report score unchanged while adding 1,450 negative candidates to the original 1,244 and retaining the same 363 positives. Report AUC against the added negatives is 0.889, compared with 0.322 against the original negatives. The pooled AUC is therefore the weighted average

$$\begin{aligned} \text{AUC}_{\text{expanded}} &= \frac{1244}{2694} \text{AUC}_{\text{original}} \\ &\quad + \frac{1450}{2694} \text{AUC}_{\text{added}} \quad (9) \\ &= 0.627. \end{aligned}$$

The added comparisons, which account for 53.8% of all positive–negative pairs, raise AUC without changing the ordering of the original candidates. At a two-concept budget, the report continues to prioritize passage content: answer accuracy is 41.0%, below SDA’s accuracy (App. E.1; Table 13).

F.3 Importance wordings

Five matched wordings keep the episode, candidates and gold utility labels fixed. They ask about passage-only importance, remove that restriction,

Table 15: Within-episode admission ranking on the standard pool, Qwen3-8B. AUC differences are relative to the importance report; retention (%) measures whether the two admitted concepts include the useful concept.

Admission score	Within-episode AUC	Δ vs. report	Useful concept retained ($k = 2$)
Importance report	0.326	<i>reference</i>	29.8
1–10 importance rating (Park et al., 2023)	0.608	+0.282	61.2
Workspace readout	0.547	+0.221	51.0
Output likelihood	0.619	+0.293	59.2
SDA (seed 0)	0.682	+0.356	68.6
SDA (seed 1)	0.679	+0.353	69.7
SDA (seed 2)	0.688	+0.362	69.2

Table 16: Useful-concept ranking among unstated candidates, pool with unstated distractors. Qualifying episodes (345) average 5.01 candidates, including unstated useful and negative concepts. Entries: pooled AUC; $W - V$: workspace readout minus importance report.

Model	Importance report	Workspace readout	$W - V$
Qwen3-8B	0.881	0.639	−0.242
Qwen3-14B	0.880	0.654	−0.226
Qwen3-32B	0.896	0.697	−0.199
Qwen2.5-7B	0.897	0.592	−0.305
OLMo-2-7B	0.883	0.661	−0.221
Mistral-7B	0.880	0.672	−0.208

allow implied content (*implied-content wording*), state that the passage may omit the useful concept (*statedness-cued wording*), or specify a two-concept budget. The measurements cover four models on the standard pool and the pool with unstated distractors (Table 18). On Qwen3-32B, the statedness-cued wording raises standard-pool AUC from 0.398 to 0.876; the corresponding Qwen3-8B values are 0.321 and 0.537. On Mistral-7B, the implied-content wording gives 0.518, compared with 0.440 for passage-only importance. The direction and magnitude of the wording effect vary across models.

The statedness-cued wording’s ranking changes when the useful concept is stated (§6.1). Its standard-pool answer accuracy on Qwen3-8B is 32.2%, 43.0% and 46.0% at $k = 1, 2, 3$ (Table 14).

The implied-content wording permits inferences without specifying whether the passage states the useful concept, and the matched comparison trains and evaluates SDA through it. On Qwen3-8B, answer accuracy rises from 28.1%, 40.8% and 41.6% to 29.2%, 41.3% and 44.9% at $k = 1, 2, 3$. All SDA seeds exceed the wording alone at every budget; §6.4 reports the SDA results for this wording (Table 17) and the passage-only wording (Table 13).

For wording j with prompt $u_j(c, x)$, the scorer is

$$s_j(c; x) = \log \sum_{t \in Y_{\text{word}}} p_{\mathcal{M}}(t \mid u_j(c, x)) - \log \sum_{t \in N_{\text{word}}} p_{\mathcal{M}}(t \mid u_j(c, x)). \quad (10)$$

The affirmative set contains the distinct token identities of *yes*, *Yes*, their leading-space forms, and *YES*, retaining a form only when the complete form is a single token; the negative set uses the analogous forms of *no*. The uppercase forms have no leading space in this scorer. All five wordings use these same sets, the answer-position logits, and thinking-disabled prompts. The importance report in the main tables instead uses the first token of each of six case-and-space forms, as defined in App. B.6. The two aggregations account for the Mistral measurement difference in App. B.3.

Table 19 gives the exact question for each wording; it follows the passage, and *candidate* is replaced by the candidate concept.

F.4 Matched naming and position measurements

Naming the useful concept raises both teacher signals more than the importance report or the 1–10 importance rating on Qwen3-8B. Workspace, output likelihood, report and rating AUCs on the original passage are 0.543, 0.613, 0.322 and 0.568.

Table 17: Implied-content wording and distillation on the standard pool, Qwen3-8B. Answer accuracy is in percent; the final row gives answer-accuracy differences in points.

Admission policy	$k = 1$	$k = 2$	$k = 3$
Importance report	28.1	40.8	41.6
Implied-content wording	29.2	41.3	44.9
SDA with implied-content wording	37.1 ± 0.7	50.7 ± 0.7	51.0 ± 0.3
SDA minus implied-content wording	+7.9	+9.4	+6.1

Table 18: Importance wording controls on the standard pool and the pool with unstated distractors. Entries are pooled useful-concept AUCs from the untrained models’ importance-report log odds.

Model	Importance wording	Standard pool	Pool with unstated distractors
Qwen3-8B	Passage only	0.321	0.628
	Unrestricted	0.391	0.663
	Implied-content wording	0.462	0.692
	Statedness-cued wording	0.537	0.726
	Two-concept budget	0.235	0.591
Qwen3-32B	Passage only	0.398	0.671
	Unrestricted	0.555	0.749
	Implied-content wording	0.656	0.802
	Statedness-cued wording	0.876	0.892
	Two-concept budget	0.327	0.633
OLMo-2-7B	Passage only	0.327	0.630
	Unrestricted	0.358	0.643
	Implied-content wording	0.420	0.675
	Statedness-cued wording	0.547	0.721
	Two-concept budget	0.301	0.614
Mistral-7B	Passage only	0.440	0.680
	Unrestricted	0.462	0.684
	Implied-content wording	0.518	0.716
	Statedness-cued wording	0.394	0.639
	Two-concept budget	0.336	0.626

Table 19: Five importance wordings, each following the passage.

Wording	Question
Passage only	Based only on the passage above, is the concept <i>candidate</i> one of the most important things to remember in order to answer possible future questions? Answer with a single word: yes or no.
Unrestricted	Is the concept <i>candidate</i> one of the most important things to remember in order to answer possible future questions about the passage above? Answer with a single word: yes or no.
Implied-content wording	Considering both what the passage above states and what it implies without stating, is the concept <i>candidate</i> one of the most important things to remember in order to answer possible future questions? Answer with a single word: yes or no.
Statedness-cued wording	Would storing the concept <i>candidate</i> help answer a likely future question about the passage above, even if the passage never mentions it? Answer with a single word: yes or no.
Two-concept budget	A memory system can keep only two concepts from the passage above. Should the concept <i>candidate</i> be one of them, so that a later question can be answered? Answer with a single word: yes or no.

Appending a sentence naming the useful concept (the named condition) raises them to 0.952, 0.859, 0.515 and 0.581; placing the same sentence before the passage yields 0.916, 0.850, 0.582 and 0.655. Naming a distractor instead yields 0.393, 0.520, 0.249 and 0.519. Table 20 gives the six-model results summarized in §6.1. The six individual report

changes include Mistral-7B’s change from 0.411 to 0.395.

App. E.1 reports answer accuracy on this control.

Table 20: Stated-bridge control on the standard pool: original passage versus passage with one added sentence naming the useful concept. Entries: pooled AUC; *W*: workspace readout; *LL*: output likelihood; *Report*: importance report; *Rating*: 1–10 importance rating.

Model	<i>W</i>		<i>LL</i>		<i>Report</i>		<i>Rating</i>	
	Original	Named	Original	Named	Original	Named	Original	Named
Qwen3-8B	0.543	0.952	0.613	0.859	0.322	0.515	0.568	0.581
Qwen3-14B	0.571	0.954	0.636	0.864	0.355	0.607	0.565	0.591
Qwen3-32B	0.603	0.979	0.654	0.878	0.398	0.638	0.530	0.510
Qwen2.5-7B	0.520	0.879	0.600	0.890	0.427	0.456	0.481	0.481
OLMo-2-7B	0.561	0.986	0.633	0.911	0.327	0.410	0.533	0.585
Mistral-7B	0.581	0.990	0.630	0.854	0.411	0.395	0.608	0.601
Mean	0.563	0.956	0.628	0.876	0.373	0.504	0.548	0.558

G Extended Ranking Measurements

G.1 Teacher signals and passage replacement

The report ranks useful concepts above the workspace readout on Explicit; the ordering reverses on Decoupled across all eight models (Table 22). Passage replacement preserves candidates and labels (Table 21). All report measurements use Eq. (5) and the importance prompt in App. B.6.

Passage replacement across signals. Passage replacement lowers workspace AUC across all six models, including from 0.543 to 0.433 on Qwen3-8B and from 0.581 to 0.423 on Mistral-7B (Table 21). Qwen3-8B output likelihood AUC changes from 0.613 to 0.604 and report AUC from 0.322 to 0.485. Output likelihood retains a context-independent candidate prior, while workspace ranking and the report’s preference for stated candidates depend on the episode’s own passage. Replacing each passage with another episode’s also lowers SDA’s within-episode AUC from 0.682 (primary seed, own passage) to 0.540, 0.549 and 0.542 across the three seeds (pooled AUC 0.526, 0.555 and 0.534).

G.2 Breadth across checkpoints, families and stages

Across the 23-checkpoint, four-set grid in Table 22, workspace AUC exceeds report AUC on Decoupled and Compositional at every checkpoint with a defined report, including base and post-trained stages and every released OLMo-2 post-training stage measured at 1B and 7B (supervised fine-tuning (SFT), direct preference optimization (DPO) where available, and reinforcement learning with verifiable rewards (RLVR); Table 22). An entry of “—” means that the tokenizer ships no chat template, so the importance report is undefined there.

Workspace AUC exceeds report AUC at every measured Qwen3 size on Decoupled, with non-monotonic paired changes between adjacent sizes (Table 23). From 8B to 32B, the paired change is -0.0536 .

G.3 Alternate-generator and entity-bridge sets

The report’s preference for stated candidates persists with a different generator. Across six models on the alternate-generator set, report AUC ranges from 0.220 to 0.384, alongside statedness AUC 0.650–0.803 (Table 24). These measurements use the same report definition as the primary sets. The alternate-generator set also supplies SDA training episodes, so these untrained-model measurements replicate the diagnostic rather than evaluate SDA.

Entity-bridge set. On the entity-bridge set, the direction of the report’s statedness preference changes with candidate construction, while the report depends on each episode’s own passage in both regimes. Qwen2.5-7B and Qwen3-8B report AUCs reach 0.881 and 0.719, above workspace AUCs of 0.664 and 0.642 (Table 24). Both reports prefer unstated candidates: their statedness AUCs are 0.133 and 0.289. Passage replacement reduces report AUC to 0.282 and 0.340. Qwen2.5-0.5B retains the usual preference for stated candidates, with statedness AUC 0.734 and report AUC 0.270.

G.4 Admission after distillation across scales

SDA raises pooled AUC on the development sets at every measured Qwen3 scale and seed (Table 25). The measured improvement is concentrated in the admission score, while the largest absolute workspace AUC change is 0.0046.

Table 21: Passage replacement on the standard pool. Entries are pooled AUCs with the episode’s own passage or another episode’s passage; candidates and labels stay fixed.

Model	Workspace readout		Output likelihood		Importance report	
	Own	Replaced	Own	Replaced	Own	Replaced
Qwen3-8B	0.543	0.433	0.613	0.604	0.322	0.485
Qwen3-14B	0.571	0.473	0.636	0.625	0.355	0.404
Qwen3-32B	0.602	0.479	0.654	0.623	0.398	0.416
Qwen2.5-7B	0.520	0.509	0.600	0.610	0.427	0.441
OLMo-2-7B	0.561	0.477	0.633	0.617	0.327	0.412
Mistral-7B	0.581	0.423	0.630	0.620	0.411	0.455

Table 22: Workspace readout and importance report for 23 checkpoints across four sets. Entries are pooled AUCs: W , workspace readout; V , importance report. Dashes: V undefined without a chat template; gray rows: base checkpoints; OLMo-2 Instruct: RLVR stage.

Checkpoint	Explicit		Implied-answer		Checkpoint	Decoupled		Compositional	
	W	V	W	V		W	V	W	V
Qwen2.5-0.5B	.5235	.4465	.5367	.4341	Qwen2.5-0.5B	.5019	.4798	.4293	.3715
Qwen2.5-0.5B-Instruct	.5211	.5496	.5158	.2916	Qwen2.5-0.5B-Instruct	.5261	.2841	.4190	.1776
Qwen2.5-1.5B	.5100	.5006	.5816	.4160	Qwen2.5-1.5B	.5903	.4052	.5451	.3879
Qwen2.5-1.5B-Instruct	.5139	.6136	.5769	.5234	Qwen2.5-1.5B-Instruct	.5768	.4637	.5489	.2936
Qwen2.5-3B	.5369	.5621	.6147	.4560	Qwen2.5-3B	.6062	.3784	.5208	.2444
Qwen2.5-3B-Instruct	.5331	.6446	.5781	.4615	Qwen2.5-3B-Instruct	.5951	.2586	.4971	.1951
Qwen2.5-7B	.5419	.5973	.6208	.4652	Qwen2.5-7B	.6121	.3730	.5152	.2771
Qwen2.5-7B-Instruct	.5376	.6792	.6341	.5856	Qwen2.5-7B-Instruct	.6423	.4897	.5156	.3472
Qwen3-0.6B	.4932	.5306	.5472	.3264	Qwen3-0.6B	.5364	.2248	.4693	.1123
Qwen3-1.7B	.4818	.5917	.5409	.3895	Qwen3-1.7B	.5607	.1961	.4162	.1812
Qwen3-4B	.5090	.6496	.5744	.4798	Qwen3-4B	.5291	.3967	.5050	.2600
Qwen3-8B	.5090	.7100	.6204	.4534	Qwen3-8B	.6545	.3872	.5466	.2079
Qwen3-14B	.5083	.7400	.6390	.5219	Qwen3-14B	.6677	.4271	.5086	.2163
Qwen3-32B	.5084	.7480	.6139	.5511	Qwen3-32B	.6919	.4782	.5532	.2774
OLMo-2-1B	.5006	—	.6308	—	OLMo-2-1B	.6001	—	.5114	—
OLMo-2-1B-SFT	.5409	.5252	.5969	.5016	OLMo-2-1B-SFT	.5725	.3716	.4760	.3064
OLMo-2-1B-DPO	.5374	.5563	.5759	.4973	OLMo-2-1B-DPO	.5802	.3799	.4764	.2786
OLMo-2-1B-Instruct	.5354	.5551	.5795	.4899	OLMo-2-1B-Instruct	.5801	.3284	.4783	.2548
OLMo-2-7B	.4806	—	.6735	—	OLMo-2-7B	.6835	—	.5453	—
OLMo-2-7B-SFT	.4959	.5827	.6827	.5369	OLMo-2-7B-SFT	.6637	.3317	.5312	.2924
OLMo-2-7B-Instruct	.4938	.6231	.6688	.5398	OLMo-2-7B-Instruct	.6664	.3529	.5341	.2944
Mistral-7B-Instruct-v0.3	.5324	.6191	.6687	.5777	Mistral-7B-Instruct-v0.3	.6713	.4700	.5962	.3242
GPT-2	.5070	—	.5388	—	GPT-2	.5331	—	.5170	—

Table 23: Qwen3 dense model sizes on Decoupled. Upper block: workspace readout minus importance report ($W - V$); lower block: changes in $W - V$ between sizes.

Checkpoint	$W - V$
Qwen3-1.7B	+0.3646
Qwen3-4B	+0.1324
Qwen3-8B	+0.2673
Qwen3-14B	+0.2405
Qwen3-32B	+0.2137
Size change	$\Delta(W - V)$
1.7B $\rightarrow$ 4B	−0.2322
4B $\rightarrow$ 8B	+0.1349
8B $\rightarrow$ 14B	−0.0268
14B $\rightarrow$ 32B	−0.0268
8B $\rightarrow$ 32B	−0.0536

Table 24: Diagnostic replications with untrained models on the alternate-generator set (also a source of SDA training episodes) and the entity-bridge set. Pooled AUCs: W , workspace readout; V , importance report with thinking disabled; $W - V$, their difference.

Set	Model	W	V	$W - V$
Alternate-generator	Qwen2.5-7B	0.490	0.384	+0.106
	Qwen3-8B	0.515	0.220	+0.295
	Qwen3-14B	0.521	0.267	+0.254
	Qwen3-32B	0.522	0.275	+0.247
	OLMo-2-7B	0.547	0.353	+0.194
	Mistral-7B	0.507	0.294	+0.213
Entity-bridge	Qwen2.5-0.5B	0.569	0.270	+0.299
	Qwen2.5-7B	0.664	0.881	-0.217
	Qwen3-8B	0.642	0.719	-0.076

Table 25: Admission ranking on the Decoupled and Compositional development sets across Qwen3 scales. V : pooled AUC of the admission score (untrained before, SDA after); ΔV : paired change; $|\Delta W|$: largest absolute workspace readout change across seeds.

Model	Set	V before	Seed	V after	ΔV	$ \Delta W $
Qwen3-8B	Decoupled	0.387	0	0.756	+0.369	0.0033
			1	0.739	+0.352	
			2	0.758	+0.371	
Qwen3-8B	Compositional	0.208	0	0.618	+0.410	0.0021
			1	0.623	+0.415	
			2	0.619	+0.412	
Qwen3-14B	Decoupled	0.427	0	0.702	+0.275	0.0029
			1	0.669	+0.242	
			2	0.680	+0.253	
Qwen3-14B	Compositional	0.216	0	0.529	+0.312	0.0029
			1	0.458	+0.242	
			2	0.481	+0.265	
Qwen3-32B	Decoupled	0.478	0	0.773	+0.295	0.0037
			1	0.762	+0.284	
			2	0.721	+0.243	
Qwen3-32B	Compositional	0.277	0	0.622	+0.344	0.0046
			1	0.624	+0.347	
			2	0.588	+0.310	

**H Teacher and Optimization
Measurements**

**H.1 Teacher composition and workspace
readout**

Table 26 reports answer accuracy on the statedness-balanced pool for all three teacher compositions; Table 27 gives their paired differences.

Table 26: Teacher composition on the statedness-balanced pool, five models. Answer accuracy (%); the two “Without” columns show students trained with one teacher signal removed. Table 27 gives paired differences.

Model	Budget	SDA	Without the workspace readout	Without the output likelihood
Qwen3-8B	$k = 1$	40.5 ± 0.3	42.1 ± 1.2	36.7 ± 0.4
	$k = 2$	53.7 ± 1.1	54.8 ± 0.3	49.8 ± 0.8
	$k = 3$	51.8 ± 0.5	50.6 ± 0.3	50.5 ± 0.4
Qwen3-14B	$k = 1$	35.5 ± 1.1	36.2 ± 0.3	34.3 ± 1.6
	$k = 2$	44.1 ± 0.8	43.4 ± 0.3	42.4 ± 1.2
	$k = 3$	45.3 ± 0.6	45.8 ± 1.1	45.2 ± 0.6
Qwen3-32B	$k = 1$	44.9 ± 1.2	43.4 ± 1.2	45.7 ± 0.8
	$k = 2$	53.3 ± 1.1	52.2 ± 2.1	54.7 ± 0.7
	$k = 3$	54.8 ± 1.0	55.3 ± 1.5	56.6 ± 0.6
Qwen2.5-7B	$k = 1$	17.4 ± 1.3	16.2 ± 4.1	17.6 ± 2.4
	$k = 2$	33.3 ± 2.4	32.2 ± 6.5	32.4 ± 1.9
	$k = 3$	39.1 ± 1.0	37.6 ± 4.5	37.3 ± 1.3
OLMo-2-7B	$k = 1$	49.5 ± 0.9	46.2 ± 1.8	45.2 ± 3.9
	$k = 2$	54.7 ± 0.2	53.4 ± 1.5	51.1 ± 2.6
	$k = 3$	55.5 ± 0.4	55.5 ± 1.1	55.2 ± 1.0

Table 27: Teacher signal ablations on the statedness-balanced pool, five models. Entries are SDA minus students trained with one teacher signal removed, in accuracy points.

Model	Teacher	$k = 1$	$k = 2$	$k = 3$
Qwen3-8B	Without the workspace readout	−1.6	−1.1	+1.2
	Without the output likelihood	+3.8	+4.0	+1.3
Qwen3-14B	Without the workspace readout	−0.7	+0.6	−0.6
	Without the output likelihood	+1.1	+1.6	+0.1
Qwen3-32B	Without the workspace readout	+1.5	+1.0	−0.5
	Without the output likelihood	−0.8	−1.5	−1.7
Qwen2.5-7B	Without the workspace readout	+1.1	+1.1	+1.5
	Without the output likelihood	−0.3	+0.9	+1.8
OLMo-2-7B	Without the workspace readout	+3.3	+1.3	+0.0
	Without the output likelihood	+4.3	+3.6	+0.3

We construct the paired-passage set after locking the confirmation set, from its 363 episodes. Matching these episodes into 181 disjoint pairs yields 362 episodes, with each source episode in at most one pair and one episode unpaired. Each member retains its passage, future question, and answer. Its own useful concept is the only candidate with a positive gold label, and its partner’s useful concept is a negative candidate. Each member’s candidate list is the union of the two original lists, averaging 8.86 candidates per episode, with between 8 and 12 candidates. Pair members contain the same candidates in different shuffled orders; no member retains its source order. In all 362 episodes, no useful concept appears as a word in either passage of its pair. Word presence therefore cannot separate the useful concept from the partner’s useful concept; only the content of the passage can. SDA’s configuration was fixed before this set was built (App. C). Figure 6 shows one complete pair.

We evaluate Qwen3-8B on all 362 episodes; the per-model and pooled analyses use the same pairs.

Table 28 gives the paired-passage results for each model.

Table 29 reports SDA answer accuracy of 38.1%, 49.1% and 52.3% at $k = 1, 2, 3$, with 48.2% retention at $k = 1$. Removing the workspace readout from SDA’s teacher lowers retention by 5.3 points and leaves answer accuracy unchanged on this set: the difference is −0.1 points. Removing the output likelihood lowers both retention, by 2.9 points, and answer accuracy, by 3.0 points. Removing the workspace readout also lowers retention by 3.1 and 3.5 points at $k = 2, 3$. The both-of-a-pair column reports 20.8% for SDA, 17.9% for the student trained without the workspace readout and 20.4% for the student trained without the output likelihood. A ranking that depends only on candidate identity cannot raise this rate above zero at one stored concept. Against the untrained baselines at one stored

Example 3: Paired-passage set, first episode

Passage. Soraya carefully arranged the menorah on the windowsill, lighting the first candle at sunset. Her family gathered, singing traditional songs and spinning dreidels. Plates of latkes and sufganiyot were passed around. The evening was filled with laughter and storytelling, each tale passed down for generations.

Candidates, in supplied order. clients, bouquets, lilies, florist, hanukkah, song, latke, candle, holiday, vases.
Useful concept. hanukkah.

Future question. What religion does Soraya traditionally observe during this holiday?
Answer. judaism.

Example 4: Paired-passage set, second episode

Passage. Minh spent the afternoon arranging yellow and white lilies in tall crystal vases. She snipped stems and removed wilted leaves to ensure the displays would last. After consulting her chart, she delivered the finished bouquets to waiting clients preparing for a special event downtown.

Candidates, in supplied order. florist, latke, song, holiday, clients, vases, candle, bouquets, hanukkah, lilies.
Useful concept. florist.

Future question. What might Minh worry about if her workplace got too cold overnight?
Answer. freezing.

Figure 6: One pair from the paired-passage set. Both passages leave *hanukkah* and *florist* unstated and share ten candidates in different orders; the passage determines which candidate is useful.

Table 28: Paired-passage set results by model. Entries are SDA minus students trained with one teacher signal removed, in percentage points; retention indicates admission of the episode’s useful concept.

Model	Without the workspace readout			Without the output likelihood	
	Budget	Retention	Accuracy	Retention	Accuracy
Qwen3-8B	$k = 1$	+5.3	−0.1	+2.9	+3.0
	$k = 2$	+3.1	−0.2	+2.4	+3.2
	$k = 3$	+3.5	+1.7	+2.2	+2.3
Qwen3-14B	$k = 1$	+9.0	+1.4	−1.4	+0.3
	$k = 2$	+6.0	+0.0	−1.3	+1.3
	$k = 3$	+2.3	+0.6	+1.6	+1.8
Qwen3-32B	$k = 1$	+3.8	+1.6	−10.6	−1.9
	$k = 2$	+1.8	+0.7	−6.2	−0.3
	$k = 3$	+0.7	+0.2	−4.5	−0.8
Qwen2.5-7B	$k = 1$	+8.3	+3.0	−1.6	+0.8
	$k = 2$	+6.2	+1.8	+0.6	+0.6
	$k = 3$	+4.4	+1.5	+4.3	+0.4
OLMo-2-7B	$k = 1$	+10.0	+3.4	+3.9	+4.4
	$k = 2$	+6.0	+0.7	+5.0	+3.1
	$k = 3$	+4.0	+1.7	+4.8	+3.4

concept, SDA improves answer accuracy by 10.5 points over the importance report and by 2.5 points over the 1–10 importance rating.

Table 30 gives the passage-replacement measurements of Figure 4 together with the workspace readout and importance report from the untrained model.

Over the 220 teacher-labeled episodes, each containing a useful concept, the two candidates SDA’s teacher labels yes differ from those labeled yes by the teacher without the workspace readout on 43.2% of episodes. They differ from those labeled yes by the teacher without the output likelihood on 45.0%. On disagreement episodes, the two candidates SDA’s teacher labels yes include the useful

concept in 63.2% against 58.9% for the teacher without the workspace readout. For the teacher without the output likelihood, these rates are 63.6% and 48.5%. Across all 220 episodes, the rates are 60.9%, 59.1% and 54.1%, respectively.

The teacher uses each candidate’s first token (Eq. (3)). The full-candidate scoring protocol appends the candidate and reads each token at the preceding position. Each token receives its peak reciprocal rank over layers; the candidate score is the mean or minimum of these peaks. Each aggregation and the first-token workspace readout take separate maxima over surface forms.

On the standard pool with Qwen3-8B, 56% of candidates span multiple tokens. Averaging over

Table 29: Paired-passage set results, Qwen3-8B. Answer accuracy and useful-concept retention (%); final column: pairs where both members retain their own useful concept (%). Rows labeled “Without ...”: students of teachers with that signal removed.

Admission policy	Accuracy			Retention	
	$k = 1$	$k = 2$	$k = 3$	$k = 1$	Both of a pair
SDA	38.1 $\pm$ 0.3	49.1 $\pm$ 0.4	52.3 $\pm$ 0.6	48.2 $\pm$ 1.4	20.8 $\pm$ 1.1
Without the workspace readout	38.2 $\pm$ 1.4	49.3 $\pm$ 1.4	50.6 $\pm$ 1.2	42.8 $\pm$ 4.7	17.9 $\pm$ 4.6
Without the output likelihood	35.2 $\pm$ 1.1	45.9 $\pm$ 1.0	50.0 $\pm$ 0.3	45.3 $\pm$ 1.0	20.4 $\pm$ 0.6
1–10 importance rating (Park et al., 2023)	35.6	43.6	46.1	39.5	13.8
Importance report	27.6	38.7	43.4	18.8	3.9
Output likelihood used directly	26.8	39.5	40.6	21.6	3.3
Workspace readout used directly	22.6	34.0	37.3	17.1	5.0

Table 30: Passage replacement on the standard pool, Qwen3-8B. Retention (%) measures whether the single stored concept is useful; drops are own-passage retention minus retention with another episode’s passage.

Admission policy	Own passage	Replaced passage	Drop
SDA	48.3 $\pm$ 1.7	27.4 $\pm$ 0.4	+20.9
Without the workspace readout	42.8 $\pm$ 5.5	39.9 $\pm$ 0.8	+2.9
Without the output likelihood	46.6 $\pm$ 0.8	20.1 $\pm$ 1.5	+26.5
Workspace readout	26.2	14.3	+11.8
Importance report	16.0	20.4	−4.4

all candidate tokens improves ranking over the matched first-token workspace readout (Table 31).

Teacher composition on the standard pool. On the standard pool (Table 32), the student trained without the workspace readout reaches 50.3%, a paired +0.3 percentage points relative to SDA; the student trained without the output likelihood reaches 47.9%, a paired −2.1 points.

Teacher scores used directly. Each teacher score used directly yields lower answer accuracy than SDA on the standard pool at $k = 2$. The combined teacher reaches 45.7%, 4.3 points below SDA; the workspace readout reaches 44.6%, 5.4 points below; and output likelihood reaches 48.8%, 1.3 points below.

H.2 Training objective

The objective baseline trains the admission score directly by applying binary cross-entropy to the yes/no log-odds in Eq. (5). Teacher labels, training episodes, adapter shape and the 414-step budget are fixed. Within-episode AUC on the validation split starts at 0.657 for the untrained model and reaches 0.824 at the final step under SDA’s vocabulary-normalized token objective. At learning rate 10^{-5} , the log-odds objective ends at 0.250, 0.333 and 0.301 across the three seeds. Validation selects the untrained checkpoint in all three runs (Table 33).

At a step size five times smaller (2×10^{-6}), validation selects a trained checkpoint after 100 steps, with within-episode AUC 0.690 and standard-pool answer accuracy of 40.2% at $k = 2$, against 50.0% for SDA’s vocabulary-normalized token prediction. At 5×10^{-6} , selection returns the untrained checkpoint. Each smaller-step-size run uses one seed. SDA’s token objective yields 9.8 points higher answer accuracy than the trained log-odds checkpoint and 9.6 points higher than the untrained checkpoint selected at the other learning rates.

Accuracy reward. SDA exceeds accuracy-reward training by 3.7 points on the standard pool; accuracy-reward refinement after SDA adds 2.8 points. These ablations use the same training episodes and adapter shape with a matched step budget (App. B.7).

I Prompts

The prompts below give the instructions for the evaluated models, the generator and the judge. Prompt 1 is the prose transcription described in App. A.2; instructions for unstated distractors and the judge’s output format are described in App. A.4 and App. D.1, respectively. Italic phrases stand for the supplied passage, candidate, memory list or question.

Prompt 1: Episode generation

Design a batch of diverse test episodes about silent bridge reasoning for a memory experiment. Each episode has a passage of four to eight sentences about a named person; invent a distinctive first name, different in every episode, and vary the domain widely. A question asked later, out of context, refers to the episode only through the person's name and requires an unstated bridge fact inferable from the passage, followed by common general knowledge. Supply a short correct answer of one to three words and five to seven candidate concepts, each a common lowercase word with a utility category and a short reason.

Exactly one candidate is the useful bridge concept, and its word must be absent from the passage. The question must share no passage content word except the person's name: someone who never read the passage must be unable to answer it. For a passage about Elena wandering the Plaza Mayor, ask what language Elena should have brushed up on for her trip, without naming the plaza in the question. The answer must differ from every candidate and be absent from the passage; combining the bridge with general knowledge yields the answer. For example, the Plaza Mayor cues *madrid*, which supports *spanish*. A reader remembering only the bridge plus the question must be able to answer.

Include two or three vivid, emotional or surprising distractors that occur literally in the passage but are useless for the question, and one or two neutral concepts that occur in the passage. Every candidate must be a single common English lowercase word. Do not reuse concepts or person names across episodes.

Prompt 2: Importance prompt, shared by SDA and the report

Passage

Based only on the passage above, is the concept "*candidate concept*" one of the most important things to remember in order to answer possible future questions? Answer with a single word: yes or no.

Prompt 3: 1–10 importance rating

Passage

On a scale of 1 to 10, where 1 is purely mundane and 10 is extremely important, rate how important it is to remember the concept "*candidate concept*" in order to answer possible future questions about this passage. Answer with a single number.

Prompt 4: ConsistencyGate, sampled form

You are a fact-checking assistant. Given a source context and a candidate fact, rate how strongly the fact is supported by the context.

Source context: *passage*

Candidate fact: *candidate concept*

Question: How strongly is this candidate fact supported by the source context above? Do not use outside knowledge.

0.0 = completely unsupported or directly contradicted by the context

0.5 = partially supported or ambiguous

1.0 = fully and directly supported by the context

Output only a single decimal number between 0.0 and 1.0.

Prompt 5: ConsistencyGate, single-pass form

You are a fact-checking assistant. Given a source context and a candidate fact, decide whether the fact is supported by the context.

Source context: *passage*

Candidate fact: *candidate concept*

Question: Is this candidate fact supported by the source context above? Do not use outside knowledge.

Answer with a single word: yes or no.

Table 31: Workspace readout variants on the standard pool, Qwen3-8B, with 1,607 candidates. Entries are pooled and within-episode AUCs.

Workspace readout	Pooled AUC	Within-episode AUC
First token	0.530	0.536
Mean across all tokens	0.574	0.571
Minimum across all tokens	0.531	0.538

Table 32: SDA ablations on the standard pool, Qwen3-8B, $k = 2$. Answer accuracy is in percent; Δ is the accuracy difference from SDA in percentage points.

Variant	Accuracy	Δ vs. SDA
SDA	50.0 ± 1.4	<i>reference</i>
<i>Students of teachers with one signal removed</i>		
Without the output likelihood	47.9 ± 0.6	-2.1
Without the workspace readout	50.3 ± 1.2	$+0.3$
<i>No distillation</i>		
Combined teacher used directly	45.7	-4.3
<i>Accuracy reward</i>		
Accuracy reward instead of distillation	46.4 ± 2.8	-3.7
SDA followed by accuracy-reward refinement	52.9 ± 0.5	$+2.8$

Prompt 6: A-MAC utility

passage

Consider storing the concept “*candidate concept*” in long-term memory. Is it actionable, does it support likely follow-up questions, or does it capture a persistent constraint or preference?

Answer with a single word: yes or no.

Table 33: Objective ablation on the standard pool, Qwen3-8B, $k = 2$. Answer accuracy is in percent; main rows use learning rate 10^{-5} , and each smaller-rate row uses one seed.

Training objective	Accuracy
SDA: token-level cross-entropy	50.0 ± 1.4
Yes/no log-odds cross-entropy	40.5 ± 0.0
Yes/no log-odds, learning rate 2×10^{-6}	40.2
Yes/no log-odds, learning rate 5×10^{-6}	40.5

Prompt 7: Context-only candidate extraction

Passage

A memory system can store only a few single-word concepts about the passage above. List up to 8 concepts that could be worth remembering for possible future questions, including things the passage clearly implies but does not state. Answer with lowercase single words separated by commas, and nothing else.

Prompt 8: Frozen reader

Earlier you read a passage. The only things you remember from it are: *stored concepts separated by commas*. Using only what you remember, answer concisely.

Question: *future question*

Answer:

Prompt 9: Blind semantic judge

Grade short answers to memory questions. For each item decide whether ANSWER gives GOLD as its answer.

Labels:

“correct”: the answer commits to GOLD or an exact equivalent (synonym, inflection, fuller name) as its single answer. Extra explanation is fine.

“hedged”: it offers two or more alternative answers (e.g. “X or Y”), or states the answer cannot be determined, even if GOLD is mentioned.

“incorrect”: anything else, including a different answer.